MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, D TYPE, DIGITAL, KEY CODE KHR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for th acquisition of an electronic D type flip-flop module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should b obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360. by using the self-addressed Standardization Document Improvement Proposa (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.226 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KHR.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/69-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

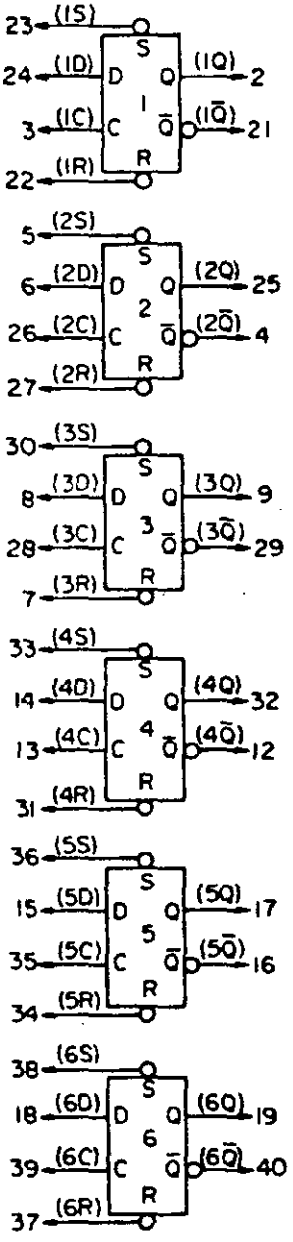
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

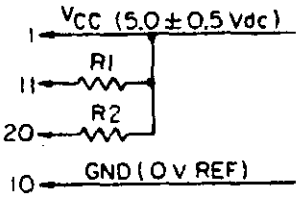
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1S-6S = Set inputs.
2. 1D-6D = Data inputs.
3. 1C-6C = Clock inputs.
4. 1R-6R = Reset inputs.
5. 1$\bar{Q}$-6$\bar{Q}$ = $\bar{Q}$ outputs.
6. 1Q-6Q = Q outputs.
7. R1 = R2 = 1.0 kΩ.

* FIGURE 1. KHR module function diagram, D type flip-flop.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B12	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C24	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D12	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
D13, D14	Output short circuit curent	VI	X	-5.1	-5.9	-5.1	-5.9	-5.1	-5.9	-5.1	-5.9	mA
E1-E12	Output leakage current	VII	X	0	1	0	1	0	1	0	1	mA
F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G36	Output voltage, low level	IX	X	0	500	0	500	0	500	0	500	mV
H1-H72	Output voltage, high level	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J6	Input current, low level	X	X	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
J7-J18	Input current, low level	X	X	-0.8	-4.4	-0.8	-4.4	-0.8	-4.4	-0.8	-4.4	mA
J19-J24	Input current, low level	X	X	-1.2	-6.6	-1.2	-6.6	-1.2	-6.6	-1.2	-6.6	mA
K1-K6	Input current, high level	XI	X	0	50	0	50	0	50	0	50	µA
K7-K18	Input current, high level	XI	X	0	100	0	100	0	100	0	100	µA
K19-K24	Input current, high level	XI	X	0	150	0	150	0	150	0	150	µA
L	Power supply current	XII	S	50	160	50	160	50	160	50	160	mA
M1-M12	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	2	17	2	17	2	20	2	20	ns
M13-M24	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	2	27	2	27	2	30	2	30	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N12	Transition time (t_{THL}), C_L = 50 pF	XIII	S	1	8	1	8	1	10	1	10	ns
N13-N24	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	2	16	2	16	2	17	2	17	ns
P1-P12	Propagation delay time (t_{PLH}), clock to output, C_L = 50 pF	XIII	S	2.0	11.5	2.0	11.5	2.0	12.5	2.0	12.5	ns
P13-P24	Propagation delay time (t_{PLH}), clock to output, C_L = 150 pF	XIII	Q	2.0	15.5	2.0	15.5	2.0	17.5	2.0	17.5	ns
Q1-Q12	Propagation delay time (t_{PHL}), clock to output, C_L = 50 pF	XIII	S	2.0	11.5	2.0	11.5	2.0	12.5	2.0	12.5	ns
Q13-Q24	Propagation delay time (t_{PHL}), clock to output, C_L = 150 pF	XIII	Q	3	18	3	18	3	19	3	19	ns
R1-R12	Propagation delay time (t_{PLH}), set and reset to output, C_L = 50 pF	XIV	S	2	8	2	8	2	9	2	9	ns
R13-R24	Propagation delay time (t_{PLH}), set and reset to output, C_L = 150 pF	XIV	Q	2.0	17.5	2.0	17.5	2.0	18.5	2.0	18.5	ns
S1-S12	Propagation delay time (t_{PHL}), set and reset to output, clock low, C_L = 50 pF	XIV	S	2.0	11.5	2.0	11.5	2.0	12.5	2.0	12.5	ns
S13-S24	Propagation delay time (t_{PHL}), set and reset to output, clock low, C_L = 150 pF	XIV	Q	2	20	2	20	2	21	2	21	ns
T1-T12	Propagation delay time (t_{PHL}), set and reset to output, clock high, C_L = 50 pF	XIV	S	2.0	14.5	2.0	14.5	2.0	15.5	2.0	15.5	ns
T13-T24	Propagation delay time (t_{PHL}), set and reset to output, clock high, C_L = 150 pF	XIV	Q	2	20	2	20	2	22	2	22	ns
U1-U6	Setup time (t_{SLH}, t_{SHL})	XV	S	450	550	450	550	450	550	450	550	ns
V1-V6	Hold time (t_{HLH}, t_{HHL})	XV	S	450	550	450	550	450	550	450	550	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
W1-W6	Maximum clock frequency and minimum clock pulsewidth check	XVI	S	40	48	40	48	40	48	40	48	ns
X1-X6	Reset recovery time	XVII	S	450	550	450	550	450	550	450	550	ns
Y1-Y12	Output crosstalk, low level	XVIII	Q	0	500	0	500	0	500	0	500	mV
Z1-Z12	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
AA	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
BB	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
CC1-CC24	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition BB, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/69-1 supersedes part number M28787/69. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	880 mW (outputs not loaded)
Supply voltage	5.0 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, low level	
V_{IL} = 0.0 V	
Input (D)	-2.2 mA maximum
Set or clock (S or C)	-4.4 mA maximum
Reset (R)	-6.6 mA maximum

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V_{IL} = 0.5 V	
Input (D)	-2.0 mA maximum
Set or clock (S or C)	-4.0 mA maximum
Reset (R)	-6.0 mA maximum
Input current, high level	
V_{IH} = 2.5 V	
Input (D)	50 µA maximum
Set or clock (S or C)	100 µA maximum
Reset (R)	150 µA maximum
V_{IH} = 5.5 V	
Input (D)	1.0 mA maximum
Set or clock (S or C)	1.0 mA maximum
Reset (R)	1.0 mA maximum
Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1 mA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.5 V)	-1.0 mA
Propagation delay time (t_{PHL})	
From clock to output	
C_L = 50 pF	2.0 ns minimum, 12.5 ns maximum
C_L = 150 pF	3.0 ns minimum, 19.0 ns maximum
From set or reset to output	
C_L = 50 pF	2.0 ns minimum, 15.5 ns maximum
C_L = 150 pF	2.0 ns minimum, 22.0 ns maximum
Propagation delay time (t_{PLH})	
From clock to output	
C_L = 50 pF	2.0 ns minimum, 12.5 ns maximum
C_L = 150 pF	2.0 ns minimum, 17.5 ns maximum
From set or reset to output	
C_L = 50 pF	2.0 ns minimum, 9.0 ns maximum
C_L = 150 pF	2.0 ns minimum, 18.5 ns maximum
Data input setup time (t_S)	
High level	3.0 ns minimum
Low level	3.0 ns minimum
Data input hold time (t_H)	
High level	2.0 ns minimum
Low level	2.0 ns minimum
Clock pulse width	10.0 ns minimum
Set pulse width	12.0 ns minimum
Reset pulse width	12.0 ns minimum
Reset recovery time	12.0 ns maximum
Clock frequency	50 MHz maximum

* 6.5 Module function. This module contains six independent D type flip-flops and two 1 kilohm pullup resistors.

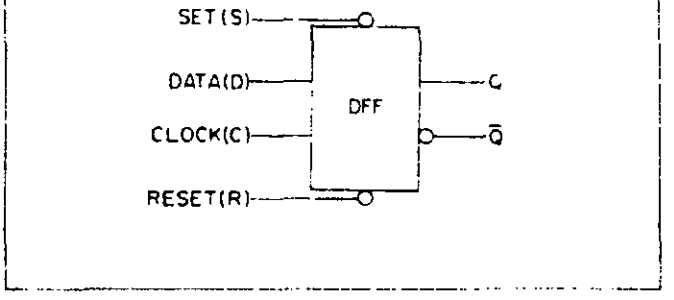
6.5.1 Input to output relationship. Each D type flip-flop has the input to output relationship of table II for positive logic.

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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

TABLE II. Input to output relationship. 1/-4/

Synchronous input before clock pulse	Required conditions of asynchronous inputs		Outputs after clock pulse goes high	
D	S	R	Q	$\overline{Q}$
X	L	H	H	L
X	H	L	L	H
L	H	H	L	H
H	H	H	H	L
X	L	L	*H	*H



1/ H = High level.
2/ L = Low level.
3/ X = Either high or low level.
4/ *H = Not stable.

6.6 Decoupling. Decoupling of test voltages and V_{CC} to minimize noise in the test setup is permissible except for test condition AA, power supply transient amplitude test.

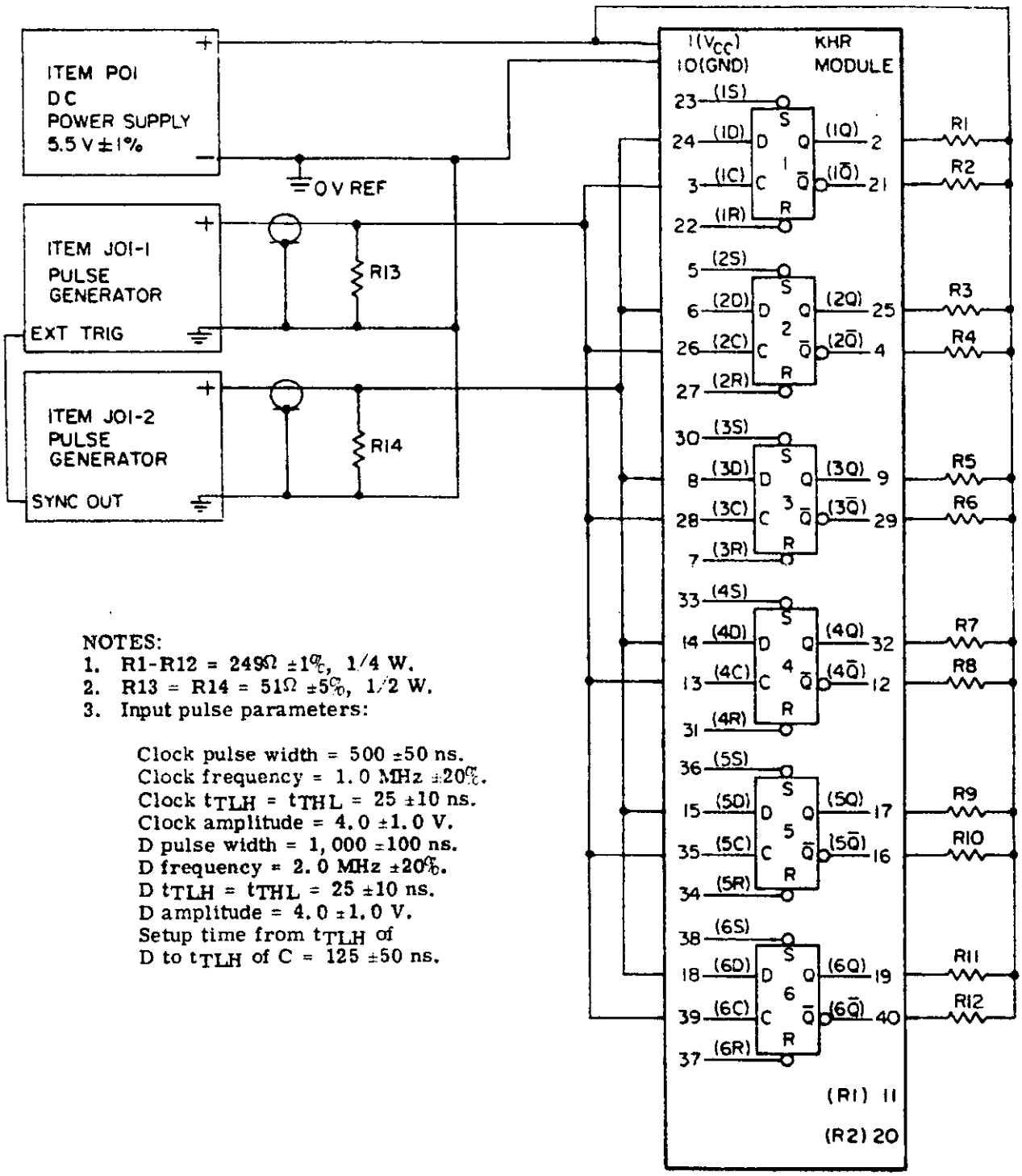
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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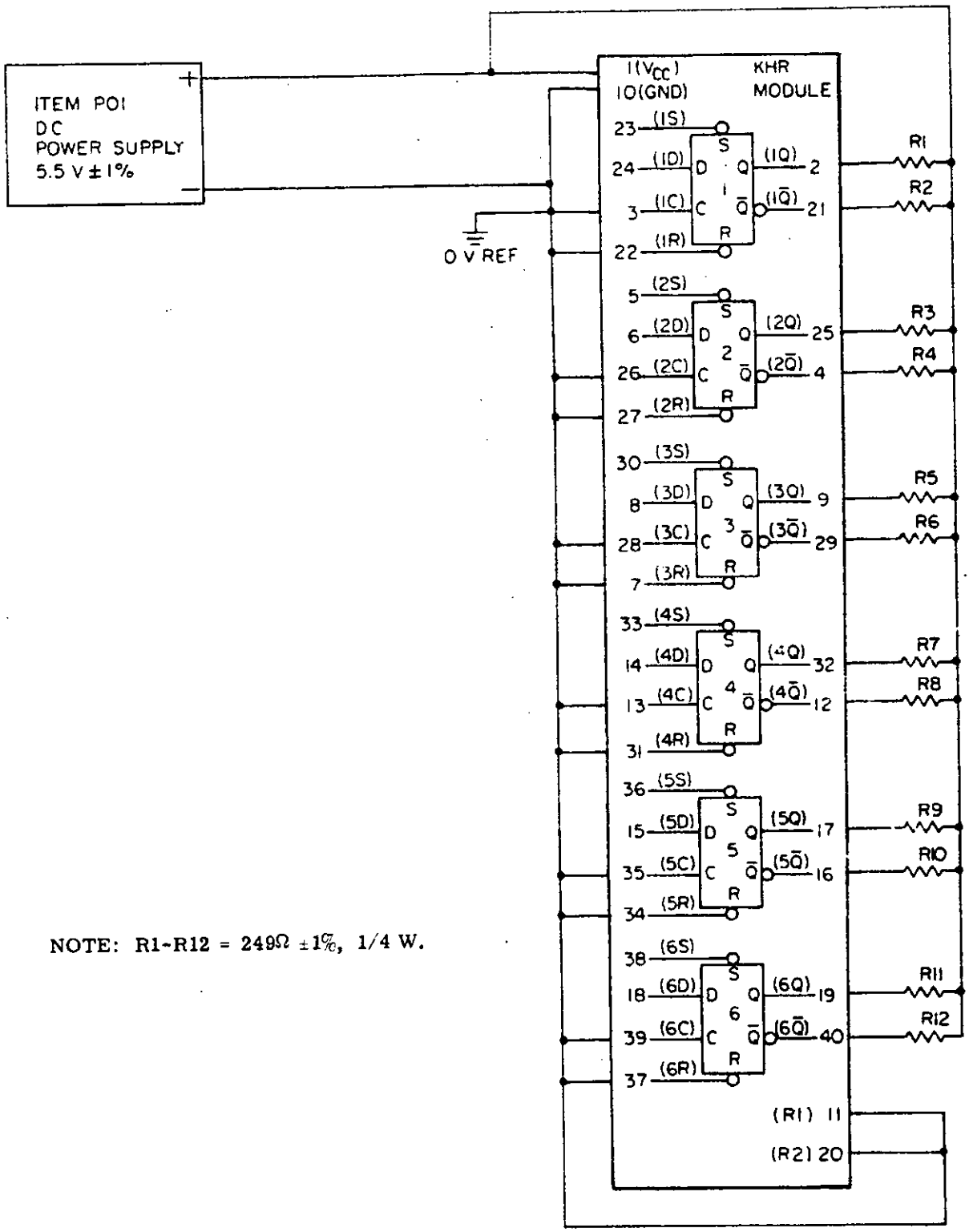


NOTES:
1. R1-R12 = 249Ω ±1%, 1/4 W.
2. R13 = R14 = 51Ω ±5%, 1/2 W.
3. Input pulse parameters:

Clock pulse width = 500 ±50 ns.
Clock frequency = 1.0 MHz ±20%.
Clock t_{TLH} = t_{THL} = 25 ±10 ns.
Clock amplitude = 4.0 ±1.0 V.
D pulse width = 1,000 ±100 ns.
D frequency = 2.0 MHz ±20%.
D t_{TLH} = t_{THL} = 25 ±10 ns.
D amplitude = 4.0 ±1.0 V.
Setup time from t_{TLH} of D to t_{TLH} of C = 125 ±50 ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R12 = 249Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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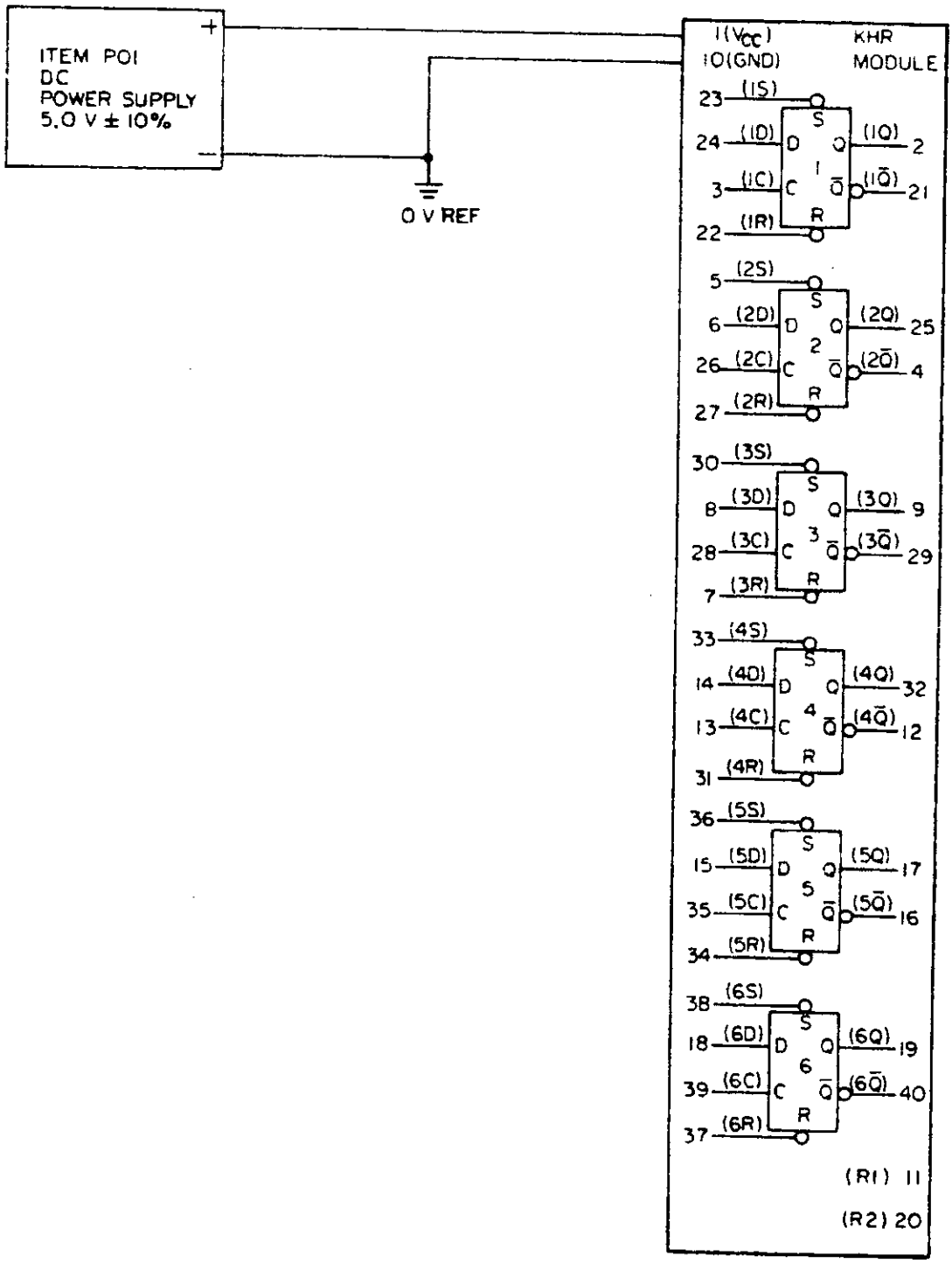


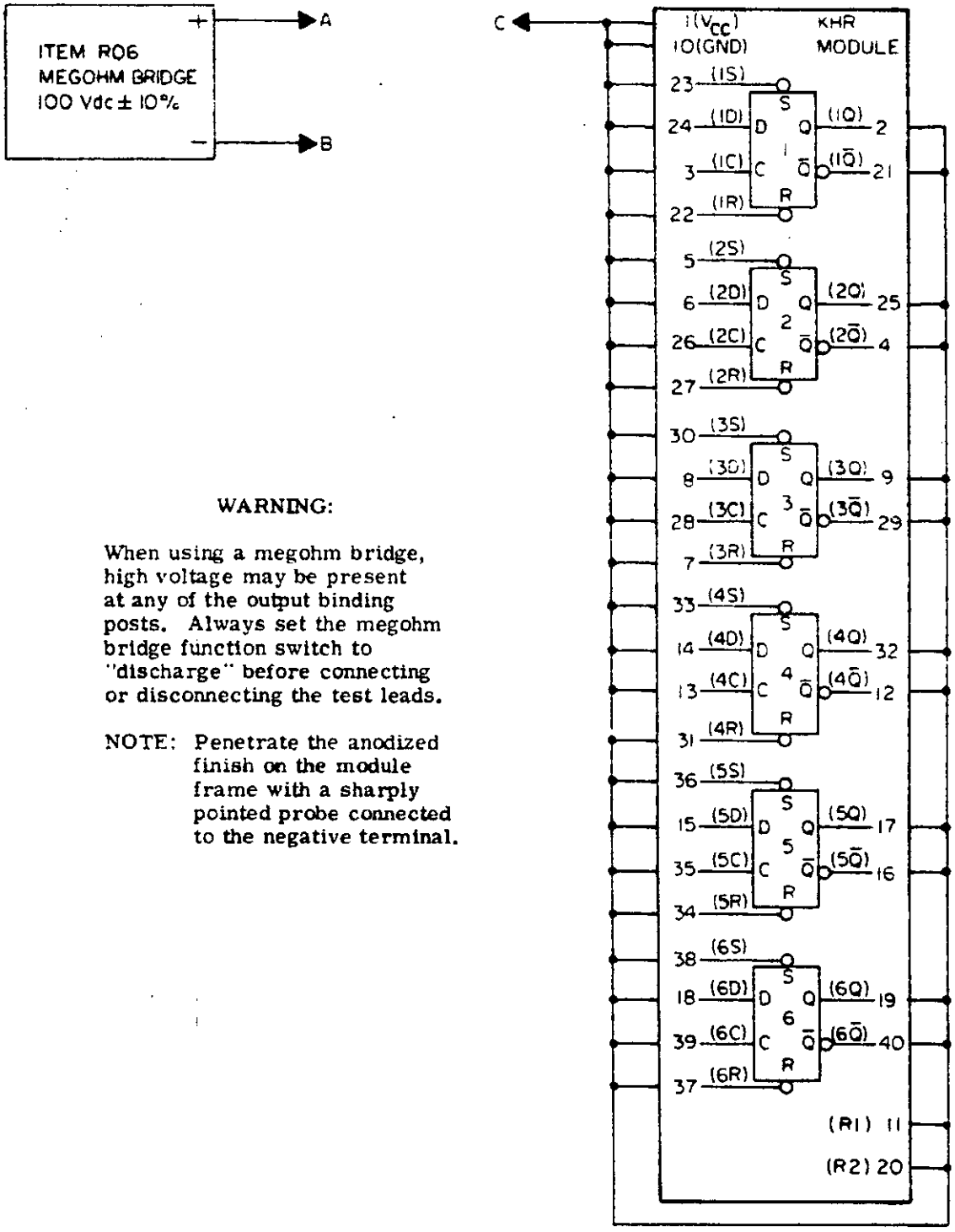
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

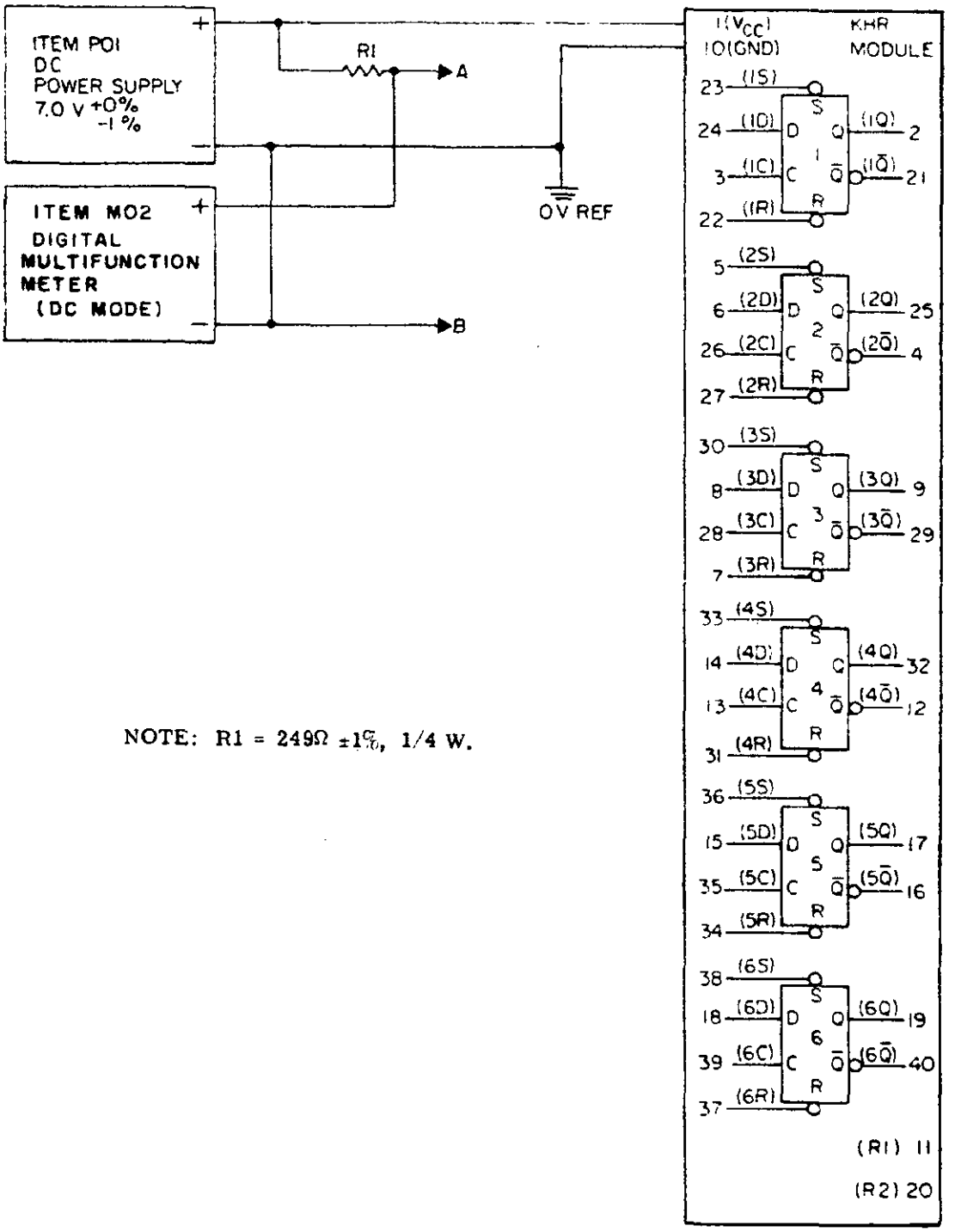
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1Q V_{OUT} (low loaded)	2	22	0.01	1.00	V	6
B2	1$\overline{Q}$	21	23				
B3	2Q	25	27				
B4	2$\overline{Q}$	4	5				
B5	3Q	9	7				
B6	3$\overline{Q}$	29	30				
B7	4Q	32	31				
B8	4$\overline{Q}$	12	33				
B9	5Q	17	34				
B10	5$\overline{Q}$	16	36				
B11	6Q	19	37				
B12	6$\overline{Q}$	40	38				

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NOTE: R1 = 249Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	1S Input breakdown current	24,22,5,6,26,27,30,8,28,7, 33,14,13,31,36,15,35, 34,38,18,39,37	23	3	0	1	mA	7
C2	1D	23,3,22,5,6,26,27,30,8, 28,7,33,14,13,31,36,15, 35,34,38,18,39,37	24	Open				
C3	1C	23,24,22,5,6,26,27,30, 8,28,7,33,14,13,31,36, 15,35,34,38,18,39,37	3	Open				
C4	1R	23,24,3,5,6,26,27,30,8, 28,7,33,14,13,31,36,15, 35,34,38,18,39,37	22	2				
C5	2S	23,24,3,22,6,27,30,8, 28,7,33,14,13,31,36,15, 35,34,38,18,39,37	5	26				
C6	2D	23,24,3,22,5,26,27,30, 8,28,7,33,14,13,31,36, 15,35,34,38,18,39,37	6	Open				
C7	2C	23,24,3,22,5,6,27,30, 8,28,7,33,14,13,31,36, 15,35,34,38,18,39,37	26	Open				
C8	2R	23,24,3,22,5,6,26,30, 8,28,7,33,14,13,31,36, 15,35,34,38,18,39,37	27	25				
C9	3S	23,24,3,22,5,6,26,27, 8,28,7,33,14,13,31,36, 15,35,34,38,18,39,37	30	25				
C10	3D	23,24,3,22,5,6,26,27, 30,28,7,33,14,13,31,36, 15,35,34,38,18,39,37	8	Open				
C11	3C	23,24,3,22,5,6,26,27, 30,8,7,33,14,13,31,36, 15,35,34,38,18,39,37	28	Open				

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TABLE V. Input breakdown current test - Continued.

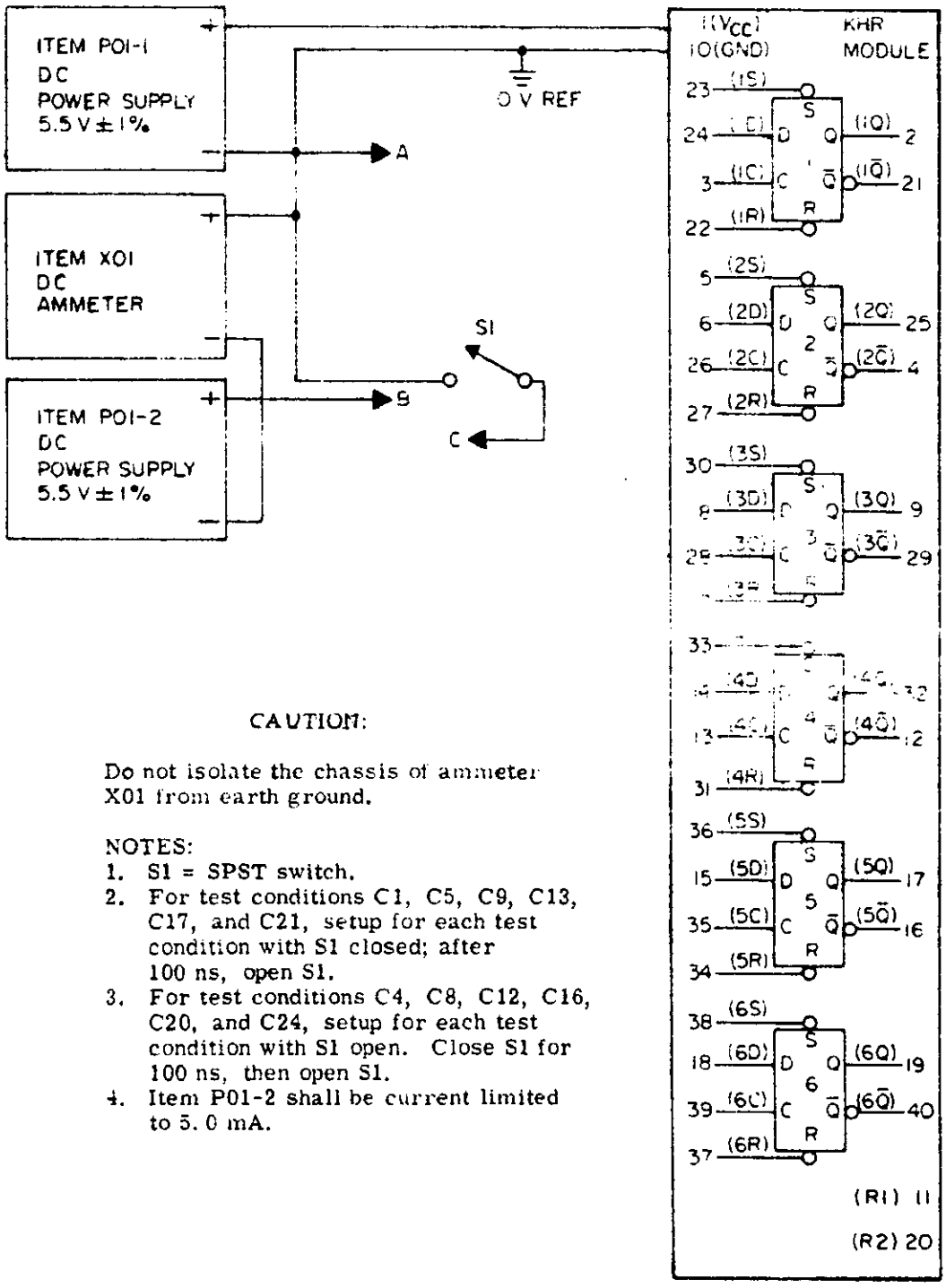
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C12	3R Input breakdown current	23,24,3,22,5,6,26,27, 30,8,28,33,14,13,31,36, 15,35,34,38,18,39,37	7	9	0	1	mA	7
C13	4S	23,24,3,22,5,6,26,27, 30,8,28,7,14,31,36,15, 35,34,38,18,39,37	33	13				
C14	4D	23,24,3,22,5,6,26,27, 30,8,28,7,33,13,31,36, 15,35,34,38,18,39,37	14	Open				
C15	4C	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,31,36, 15,35,34,38,18,39,37	13	Open				
C16	4R	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,36, 15,35,34,38,18,39,37	31	32				
C17	5S	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 15,34,38,18,39,37	36	35				
C18	5D	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,34,38,18,39,37	15	Open				
C19	5C	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,34,38,18,39,37	35	Open				
C20	5R	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,35,38,18,39,37	34	17				
C21	6S	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,35,34,18,37	38	39				
C22	6D	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,35,34,38,39,37	18	Open				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C23	6C Input breakdown current	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,35,34,38,18,37	39	Open	0	1	mA	7
C24	6R Input breakdown current	23,24,3,22,5,6,26,27, 30,8,28,7,33,14,13,31, 36,15,35,34,38,18,39	37	19	0	1	mA	7

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPST switch.
2. For test conditions C1, C5, C9, C13, C17, and C21, setup for each test condition with S1 closed; after 100 ns, open S1.
3. For test conditions C4, C8, C12, C16, C20, and C24, setup for each test condition with S1 open. Close S1 for 100 ns, then open S1.
4. Item P01-2 shall be current limited to 5.0 mA.

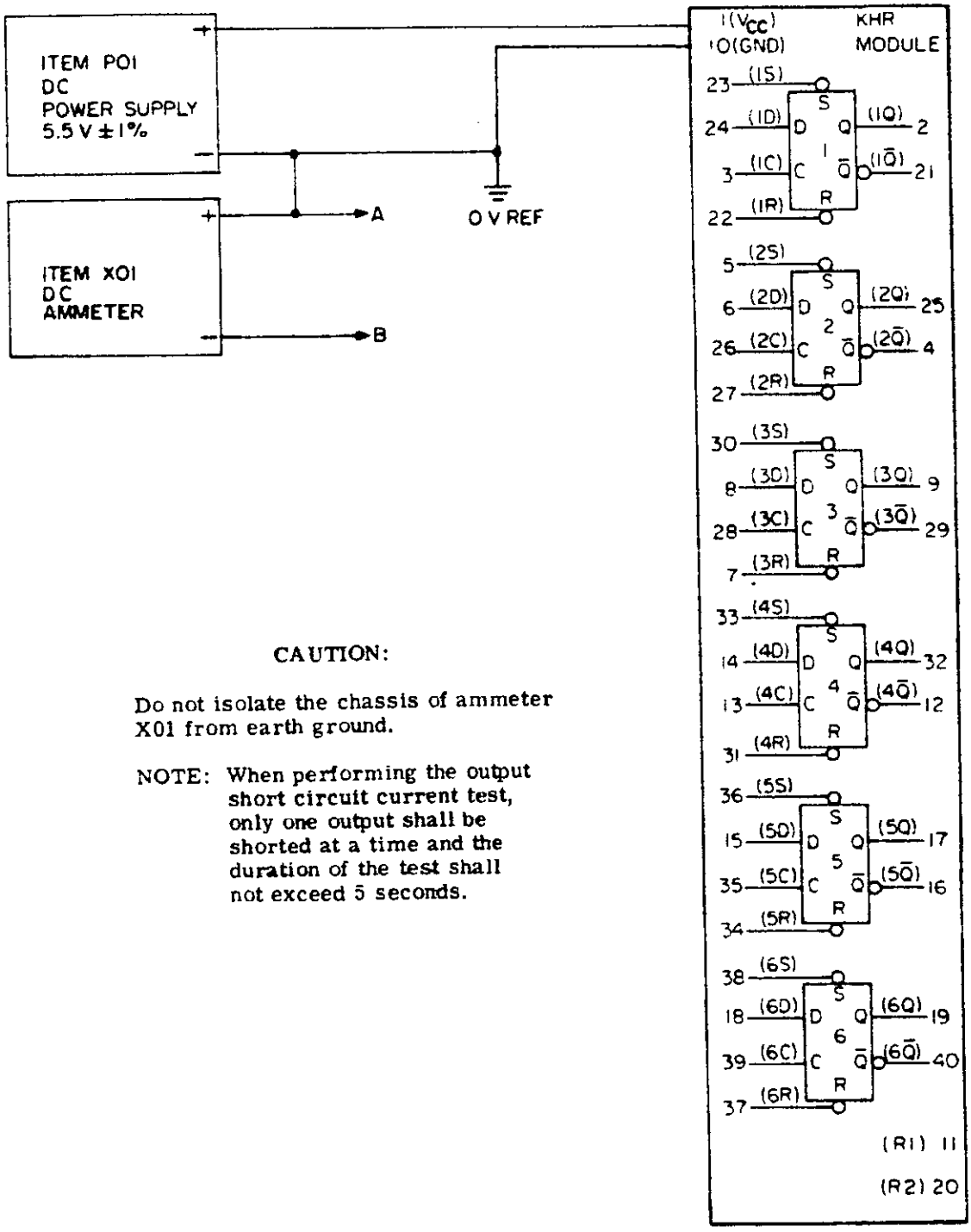
* FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1Q Output short circuit current	3,5,6,7,8,13,14,15, 18,22,23,24,26,27, 28,30,31,33,34,35, 36,37,38,39	2	-40	-100	mA	8
D2	1$\overline{Q}$		21				
D3	2Q		25				
D4	2$\overline{Q}$		4				
D5	3Q		9				
D6	3$\overline{Q}$		29				
D7	4Q		32				
D8	4$\overline{Q}$		12				
D9	5Q		17				
D10	5$\overline{Q}$		16				
D11	6Q		19				
D12	6$\overline{Q}$		40				
D13	R1		11	-5.1	-5.9		
D14	R2		20	-5.1	-5.9		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

FIGURE 8. Output short circuit current test circuit for table VI.

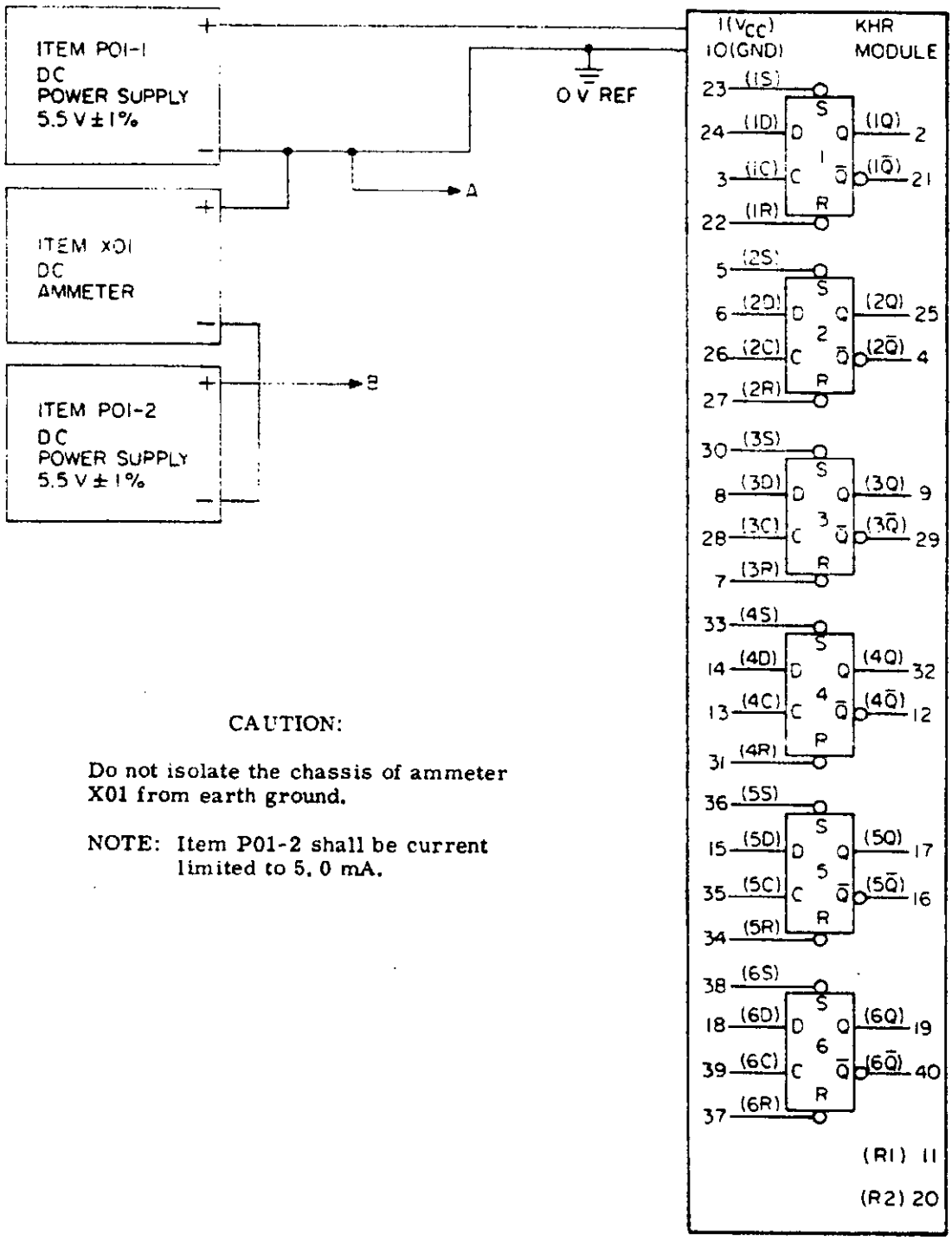
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TABLE VII. Output leakage current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1Q Output leakage current	23	2	0	1	mA	9
E2	$1\bar{Q}$	22	21				
E3	2Q	5	25				
E4	$2\bar{Q}$	27	4				
E5	3Q	30	9				
E6	$3\bar{Q}$	7	29				
E7	4Q	33	32				
E8	$4\bar{Q}$	31	12				
E9	5Q	36	17				
E10	$5\bar{Q}$	34	16				
E11	6Q	38	19				
E12	$6\bar{Q}$	37	40				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5.0 mA.

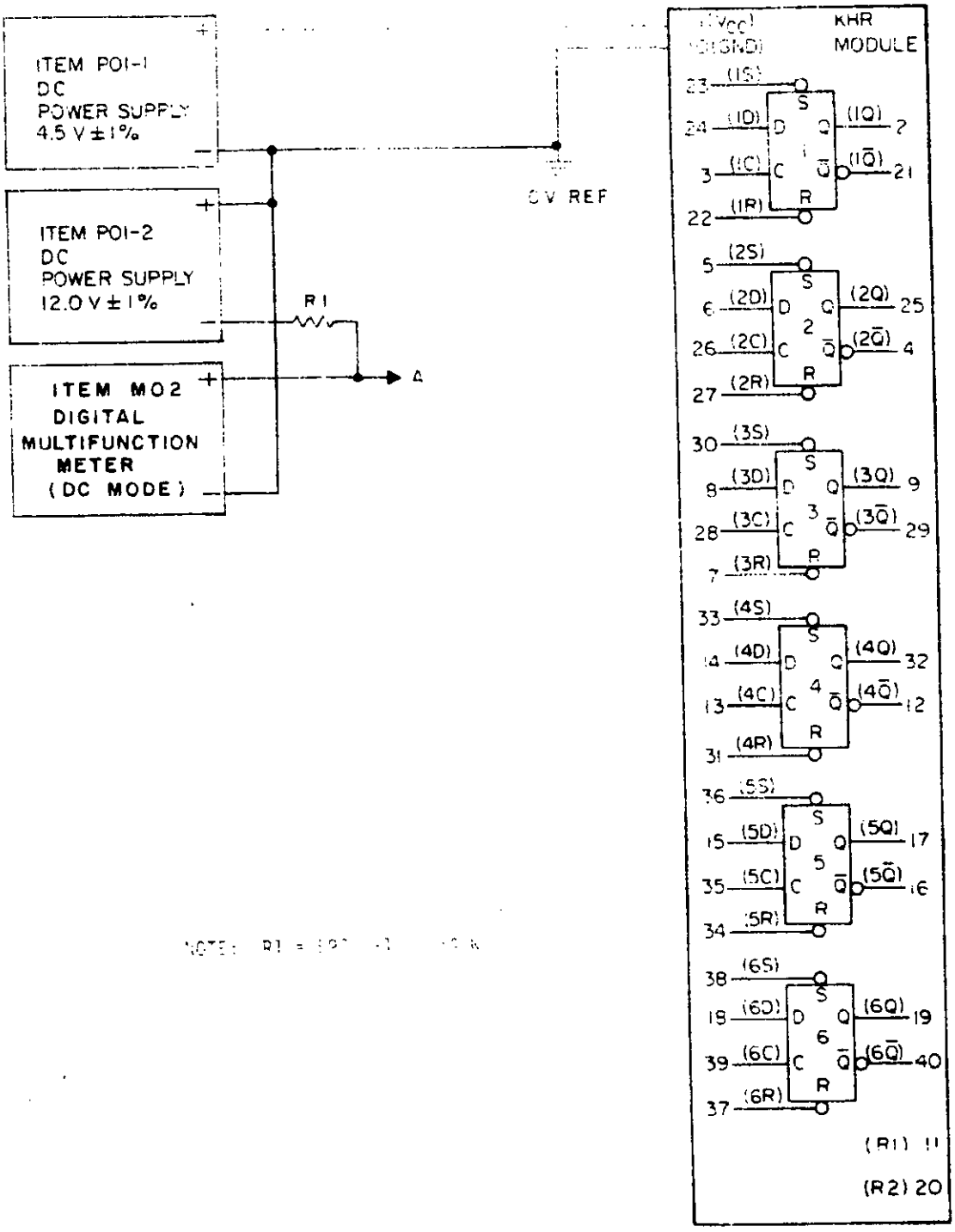
FIGURE 9. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	1S Input clamp diode voltage	23	-0.4	-1.2	V	10
F2	1D	24				
F3	1C	3				
F4	1R	22				
F5	2S	5				
F6	2D	6				
F7	2C	26				
F8	2R	27				
F9	3S	30				
F10	3D	8				
F11	3C	28				
F12	3R	7				
F13	4S	33				
F14	4D	14				
F15	4C	13				
F16	4R	31				
F17	5S	36				
F18	5D	15				
F19	5C	35				
F20	5R	34				
F21	6S	38				
F22	6D	18				
F23	6C	39				
F24	6R	37				

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NOTE: R1 = 10^ ... Ω

* FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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* TABLE IX. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q Output voltage, low level	23	22,24	3	2	0	500	mV	11
G2	1$\overline{Q}$	22	23,24		21				
G3	1Q	23,22	24		2				
G4	1Q	23,24	22		2				
G5	1$\overline{Q}$	22,24	23		21				
G6	1$\overline{Q}$	22,23,24	Open		21				
G7	2Q	5	6,27	26	25				
G8	2$\overline{Q}$	27	5,6		4				
G9	2Q	27,5	6		25				
G10	2Q	5,6	27		25				
G11	2$\overline{Q}$	6,27	5		4				
G12	2$\overline{Q}$	5,6,27	Open		4				
G13	3Q	30	8,7	28	9				
G14	3$\overline{Q}$	7	8,30		29				
G15	3Q	7,30	8		9				
G16	3Q	30,8	7		9				
G17	3$\overline{Q}$	8,7	30		29				
G18	3$\overline{Q}$	8,7,30	Open	28	29				
G19	4Q	33	14,31	13	32				
G20	4$\overline{Q}$	31	14,33	13	12				

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G21	4Q	Output voltage, low level	31,33	14	13	32	0	500	mV	11
G22	4Q		14,33	31		32				
G23	4$\overline{Q}$		14,31	33		12				
G24	4$\overline{Q}$		14,31,33	Open		12				
G25	5Q		36	15,34	35	17				
G26	5$\overline{Q}$		34	15,36		16				
G27	5$\overline{Q}$		34,36	15		17				
G28	5Q		15,36	34		17				
G29	5$\overline{Q}$		15,34	36		16				
G30	5$\overline{Q}$		15,34,36	Open		16				
G31	6Q		38	18,37	39	19				
G32	6$\overline{Q}$		37	18,38		40				
G33	6Q		37,38	18		19				
G34	6Q		18,38	37		19				
G35	6$\overline{Q}$		18,37	38		40				
G36	6$\overline{Q}$		18,37,38	Open		40				
H1	1$\overline{Q}$	Output voltage, high level	Open	23,22,24, 27,7,31, 34,37	3	21	2.5	4.5	V	
H2	1$\overline{Q}$			22,23,24, 5,30,33, 36,38		21				
H3	1Q			22,23,24, 27,7,31 34,37		2				
H4	1Q			22,23,24,5 30,33,36,38		2				

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H5	$1\overline{Q}$ Output voltage, high level	23	22,24	3	21	2.5	4.5	V	11
H6	1Q	22	24,23		2				
H7	$1\overline{Q}$	22,23	24		21				
H8	$1\overline{Q}$	24	23,22		21				
H9	1Q	24	23,22		2				
H10	$1\overline{Q}$	23,24	22		21				
H11	1Q	22,24	23		2				
H12	1Q	22,23,24	Open		2				
H13	$2\overline{Q}$	Open	5,6,27,22, 7,31,34,37	26	4				
H14	$2\overline{Q}$		5,6,27,23, 30,33,36,38		4				
H15	2Q		5,6,27,22, 7,31,34,37		25				
H16	2Q		5,6,27,23, 30,33,36,38		25				
H17	$2\overline{Q}$	5	27,6		4				
H18	2Q	27	5,6		25				
H19	$2\overline{Q}$	5,27	6		4				
H20	$2\overline{Q}$	6	5,27		4				
H21	2Q	6	5,27		25				
H22	$2\overline{Q}$	5,6	27		4				
H23	2Q	6,27	5		25				
H24	2Q	5,27,6	Open		25				
H25	$3\overline{Q}$	Open	30,8,7,22, 27,31,34,37	28	29				

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H26	3Q Output voltage, high level	Open	8,7,30,23, 5,33,36,38	28	29	2.5	4.5	V	11
H27	3Q	Open	8,7,30,22, 27,31,34,37		9				
H28	3Q	Open	7,8,30,23, 5,33,36,38		9				
H29	3$\overline{Q}$	30	7,8		29				
H30	3Q	7	8,30		9				
H31	3$\overline{Q}$	7,30	8		29				
H32	3$\overline{Q}$	8	30,7		29				
H33	3Q	8	30,7		9				
H34	3$\overline{Q}$	8,30	7		29				
H35	3Q	8,7	30		9				
H36	3Q	8,7,30	Open		9				
H37	4$\overline{Q}$	Open	33,14,31, 22,27,7, 34,37	13	12				
H38	4$\overline{Q}$		14,31,33, 23,5,30, 36,38		12				
H39	4Q		14,31,33, 22,27,7, 34,37		32				
H40	4Q		31,14,33, 23,5,30, 36,38		32				
H41	4Q	33	31,14		12				
H42	4Q	31	14,33		32				

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

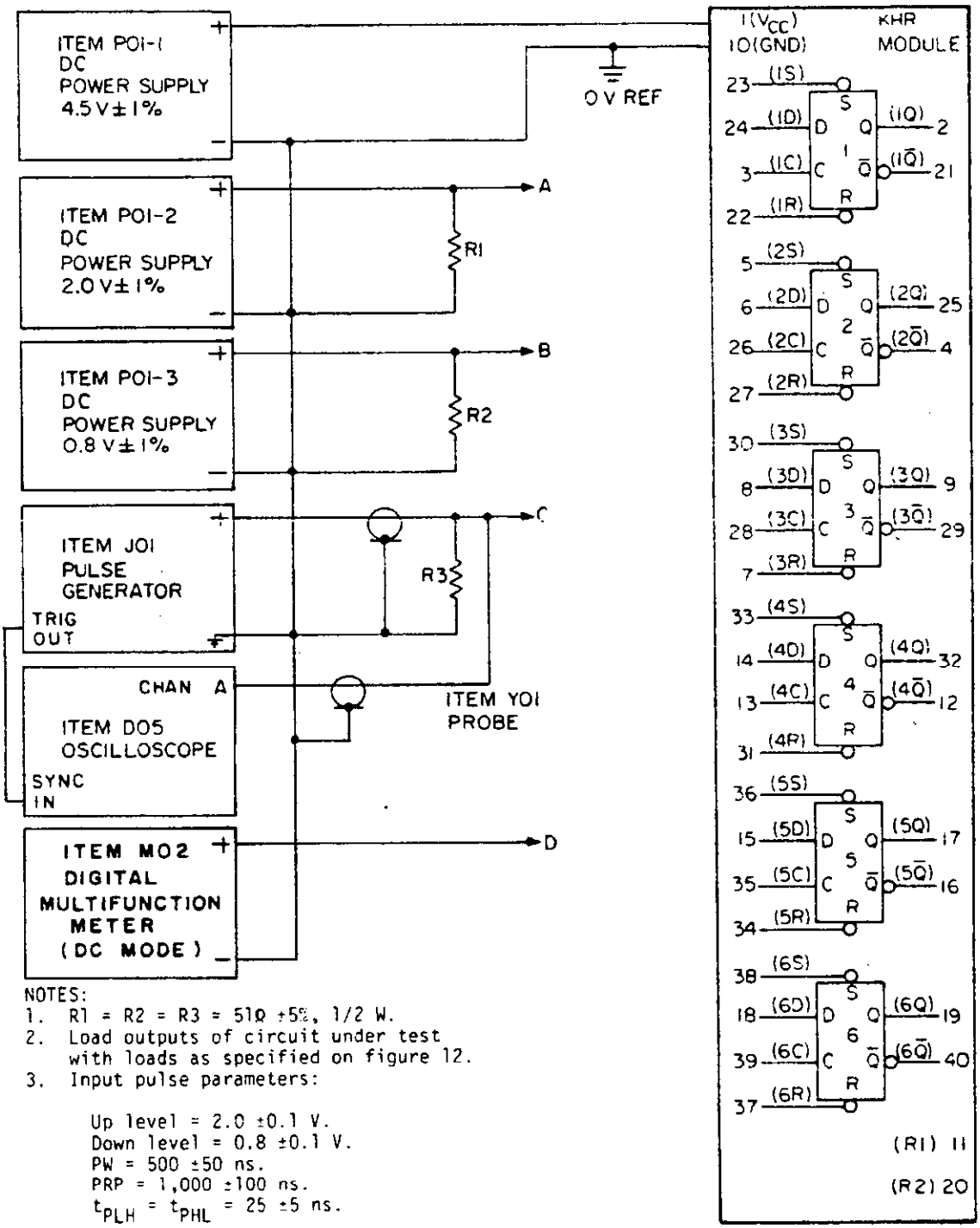
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H43	4Q̄ Output voltage, high level	31,33	14	13	12	2.5	4.5	V	11
H44	4Q̄	14	33,31		12				
H45	4Q	14	33,31		32				
H46	4Q̄	14,33	31		12				
H47	4Q	14,31	33		32				
H48	4Q	14,31,33	Open		32				
H49	5Q̄	Open	15,34,36, 22,27,7, 31,37	35	16				
H50	5Q̄		15,34,36, 23,5,30, 33,38		16				
H51	5Q		15,34,36, 22,27,7, 31,37		17				
H52	5Q		34,15,36, 23,5,30, 33,38		17				
H53	5Q̄	36	34,15		16				
H54	5Q	34	15,36		17				
H55	5Q̄	34,36	15		16				
H56	5Q̄	15	34,36		16				
H57	5Q	15	34,36		17				
H58	5Q̄	15,36	34		16				
H59	5Q	15,34	36		17				
H60	5Q	15,34,36	Open		17				
H61	6Q̄	Open	18,37,38, 22,27,7,	39	40				

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M62	6$\overline{Q}$ Output voltage, high level	Open	18,37,38, 5,23,30, 33,36	39	40	2.5	4.5	V	11
M63	6Q	Open	18,37,38, 7,22,27, 31,34		19				
M64	6Q	Open	18,38,37, 5,23,30, 33,36		19				
M65	6$\overline{Q}$	38	18,37		40				
M66	6Q	37	18,38		19				
M67	6$\overline{Q}$	37,38	18		40				
M68	6$\overline{Q}$	18	37,38		40				
M69	6Q	18	37,38		19				
M70	6$\overline{Q}$	18,38	37		40				
M71	6Q	18,37	38		19				
M72	6Q	18,37,38	Open		19				

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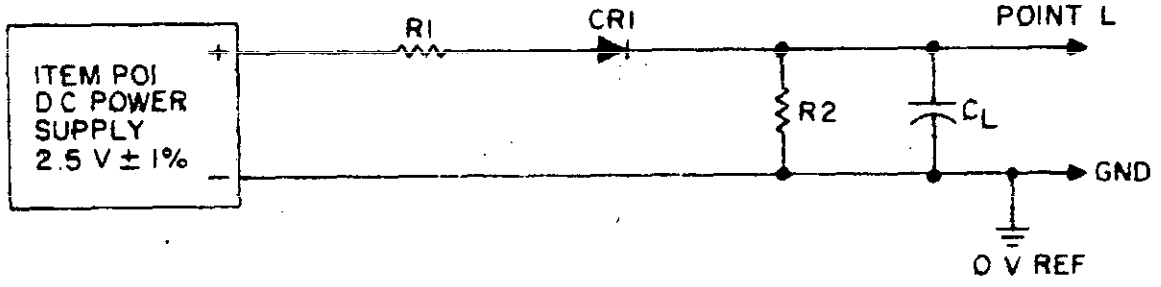
NOTES:
1. R1 = R2 = R3 = 510 ±5%, 1/2 W.
2. Load outputs of circuit under test with loads as specified on figure 12.
3. Input pulse parameters:

   Up level = 2.0 ±0.1 V.
   Down level = 0.8 ±0.1 V.
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{PLH} = t_{PHL}$ = 25 ±5 ns.

4. Set item J01 to single pulse, connect module pins per table IX, pulse item J01 once and take measurement.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select Value for a current flow of 20 mA ±5% with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. Test type S:

   C_L = 50 ±5 pF (including probe and parasitic capacitance).

   Test type Q:

   C_L = 150 ±5 pF (including probe and parasitic capacitance).

5. Capacitor C_L is not required for do electrical test conditions G1-G36 and H1-H72.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm)

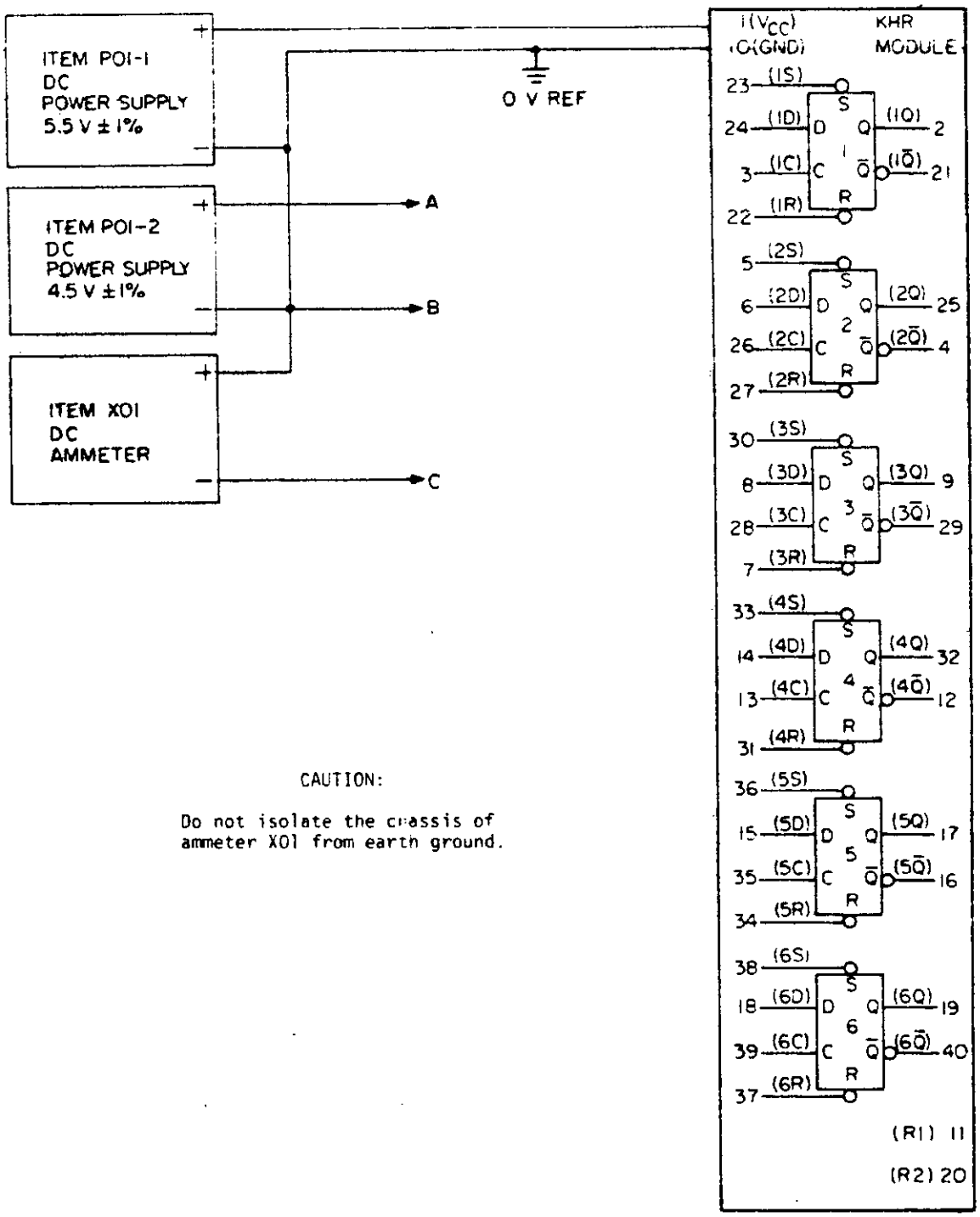
FIGURE 12. KHR logic load circuit.

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TABLE X. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1D Input current, low level	22	3	24	-0.4	-2.2	mA	13
J2	2D	27	26	6				
J3	3D	7	28	8				
J4	4D	31	13	14				
J5	5D	34	35	15				
J6	6D	37	39	18				
J7	1S	Open	22,24	23	-0.8	-4.4		
J8	2S		6,27	5				
J9	3S		7,8	30				
J10	4S		14,31	33				
J11	5S		15,34	36				
J12	6S		18,37	38				
J13	1C	22	23,24	3				
J14	2C	27	5,6	26				
J15	3C	7	8,30	28				
J16	4C	31	14,33	13				
J17	5C	34	15,36	35				
J18	6C	37	18,38	39				
J19	1R	24	3,23	22	-1.2	-6.6		
J20	2R	6	26,5	27				
J21	3R	8	28,30	7				
J22	4R	14	13,33	31				
J23	5R	15	35,36	34				
J24	6R	18	39,38	37				

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 13. Input current (low-level) test circuit for table V.

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TABLE XI. Input current (high-level) test.

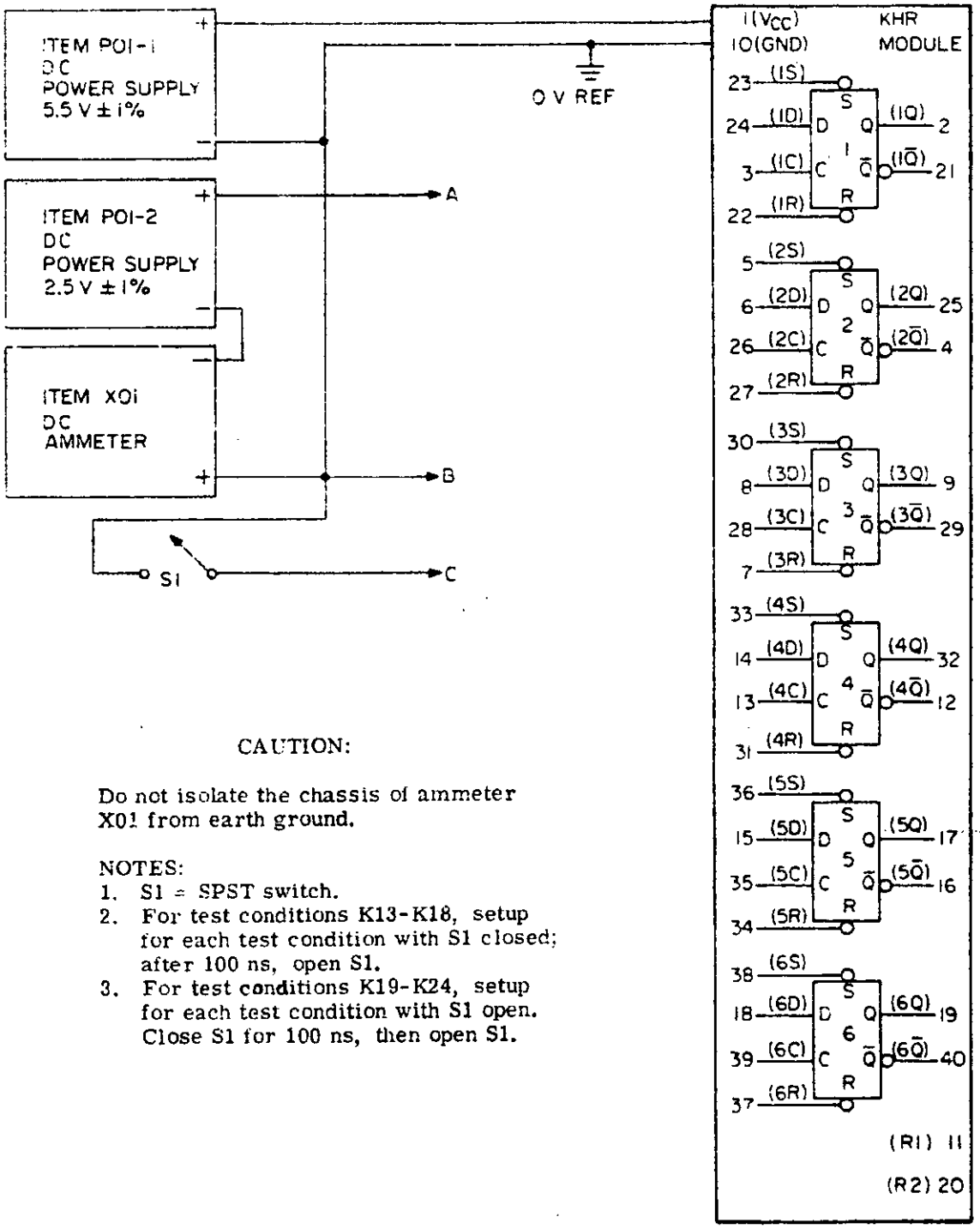
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
K1	1D Input current, high level	24	3,5,6,7,8,13,14,15,18,22,23, 26,27,28,30,31,33,34,35,36, 37,38,39	Open	0	50	µA	14
K2	2D	6	3,5,7,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,37,38,39					
K3	3D	8	3,5,6,7,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,37,38,39					
K4	4D	14	3,5,6,7,8,13,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,37,38,39					
K5	5D	15	3,5,6,7,8,13,14,18,22,23,24, 26,27,28,30,31,33,34,35,36, 37,38,39					
K6	6D	18	3,5,6,7,8,13,14,15,22,23,24, 26,27,28,30,31,33,34,35,36, 37,38,39					
K7	1C	3	5,6,7,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,37,38,39			100		
K8	2C	26	3,5,6,7,8,13,14,15,18,22,23, 24,27,28,30,31,33,34,35,36, 37,38,39					
K9	3C	28	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,30,31,33,34,35,36, 37,38,39					
K10	4C	13	3,5,6,7,8,14,15,18,22,23,24, 26,27,28,30,31,33,34,35,36, 37,38,39					
K11	5C	35	3,5,6,7,8,13,14,15,18,22,23, 24,25,27,28,30,31,33,34,36, 37,38,39					
K12	6C	39	3,5,6,7,8,13,14,15,18,22, 23,24,26,27,28,30,31,33,34, 35,36,37,38					

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TABLE IX. Input current (high level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K13	1S	Input current, high level	23	5,6,7,8,13,14,15,18,22,24, 26,27,28,30,31,33,34,35,36, 37,38,39	3	0	100	μA	14
K14	2S		5	3,6,7,8,13,14,15,18,22,23, 24,27,28,30,31,33,34,35,36, 37,38,39	26				
K15	3S		30	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,31,33,34,35,36,37, 38,39	28				
K16	4S		33	3,5,6,7,8,14,15,18,22,23,24, 26,27,28,30,31,34,35,36,37, 38,39	13				
K17	5S		36	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,37, 38,39	35				
K18	6S		38	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,37	39				
K19	1R		22	3,5,6,7,8,13,14,15,18,23,24, 26,27,28,30,31,33,34,35,36, 37,38,39	2		150		
K20	2R		27	3,5,6,7,8,13,14,15,18,22,23, 24,26,28,30,31,33,34,35,36, 37,38,39	25				
K21	3R		7	3,5,6,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,37,38,39	9				
K22	4R		31	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,28,30,33,34,35,36, 37,38,39	32				
K23	5R		34	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,35,36, 37,38,39	17				
K24	6R		37	3,5,6,7,8,13,14,15,18,22,23, 24,26,27,28,30,31,33,34,35, 36,38,39	19				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPST switch.
2. For test conditions K13-K18, setup for each test condition with S1 closed; after 100 ns, open S1.
3. For test conditions K19-K24, setup for each test condition with S1 open. Close S1 for 100 ns, then open S1.

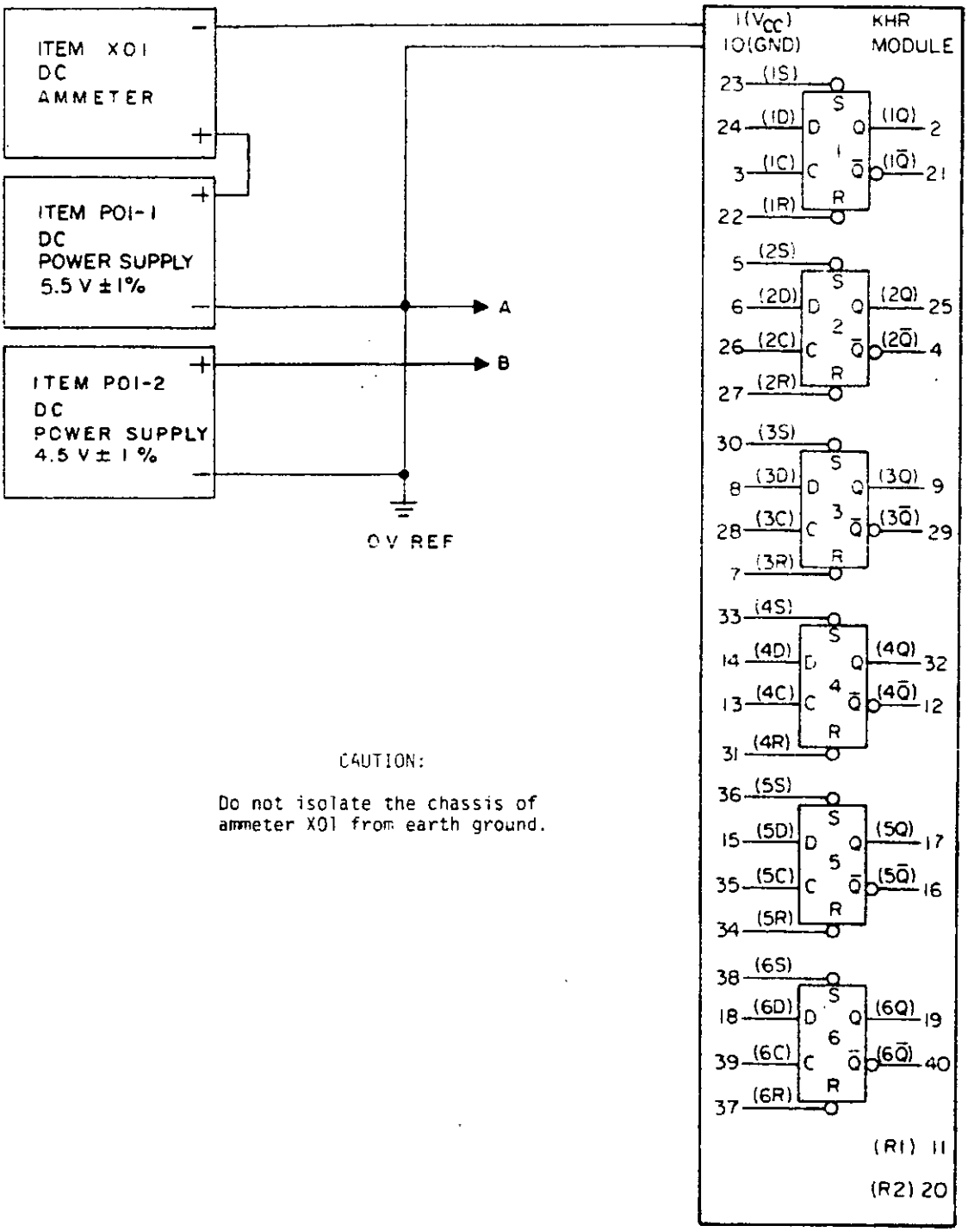
FIGURE 14. Input current (high-level) test circuit for table XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Connect		Initial limits -25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
1	Power supply current	3,13,26,28, 35,39,11,20	5,6,7,8,14, 15,16,22,23, 24,27,30,31, 33,34,36,37, 38	50	160	mA	15

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

* FIGURE 15. Power supply current test circuit for table XII.

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* TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1Q Transition time (t_{TLH}), C_L = 50 pF	3	24	2	2	17	ns	16,17
M2	1$\bar{Q}$	3	24	21				
M3	2Q	26	6	25				
M4	2$\bar{Q}$	26	6	4				
M5	3Q	28	8	9				
M6	3$\bar{Q}$	28	8	29				
M7	4Q	13	14	32				
M8	4$\bar{Q}$	13	14	12				
M9	5Q	35	15	17				
M10	5$\bar{Q}$	35	15	16				
M11	6Q	39	18	19				
M12	6$\bar{Q}$	39	18	40				
M13	1Q Transition time (t_{TLH}), C_L = 150 pF	3	24	2		27		
M14	1Q	3	24	21				
M15	2Q	26	6	25				
M16	2$\bar{Q}$	26	6	4				
M17	3Q	28	8	9				
M18	3$\bar{Q}$	28	8	29				
M19	4Q	13	14	32				
M20	4$\bar{Q}$	13	14	12				
M21	5Q	35	15	17				
M22	5$\bar{Q}$	35	15	16				
M23	6Q	39	18	19				
M24	6$\bar{Q}$	39	18	40				

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* TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1Q Transition time (t_{THL}), C_L = 50 pF	3	24	2	1	8	ns	16,17
N2	1$\overline{Q}$	3	24	21				
N3	2Q	26	6	25				
N4	2$\overline{Q}$	26	6	4				
N5	3Q	28	8	9				
N6	3$\overline{Q}$	28	8	29				
N7	4Q	13	14	32				
N8	4$\overline{Q}$	13	14	12				
N9	5Q	35	15	17				
N10	5$\overline{Q}$	35	15	16				
N11	6Q	39	18	19				
N12	6$\overline{Q}$	39	18	40				
N13	1Q Transition time (t_{THL}), C_L = 150 pF	3	24	2	2	16		
N14	1$\overline{Q}$	3	24	21				
N15	2Q	26	6	25				
N16	2$\overline{Q}$	26	6	4				
N17	3Q	28	8	9				
N18	3$\overline{Q}$	28	8	29				
N19	4Q	13	14	32				
N20	4$\overline{Q}$	13	14	12				
N21	5Q	35	15	17				
N22	5$\overline{Q}$	35	15	16				
N23	6Q	39	18	19				
N24	6$\overline{Q}$	39	18	40				

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* TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	1Q Propagation delay time (t_{PLH}), clock to output, C_L = 50 pF	3	24	2	2.0	11.5	ns	16,17
P2	1$\bar{Q}$	3	24	21				
P3	2Q	26	6	25				
P4	2$\bar{Q}$	26	6	4				
P5	3Q	28	8	9				
P6	3$\bar{Q}$	28	8	29				
P7	4Q	13	14	32				
P8	4$\bar{Q}$	13	14	12				
P9	5Q	35	15	17				
P10	5$\bar{Q}$	35	15	16				
P11	6Q	39	18	19				
P12	6$\bar{Q}$	39	19	40				
P13	1Q Propagation delay time (t_{PLH}), clock to output, C_L = 150 pF	3	24	2		15.5		
P14	1$\bar{Q}$	3	24	21				
P15	2Q	26	6	25				
P16	2$\bar{Q}$	26	6	4				
P17	3Q	28	8	9				
P18	3$\bar{Q}$	28	8	29				
P19	4Q	13	14	32				
P20	4$\bar{Q}$	13	14	12				
P21	5Q	35	15	17				
P22	5$\bar{Q}$	35	15	16				
P23	6Q	39	18	19				
P24	6$\bar{Q}$	39	18	40				

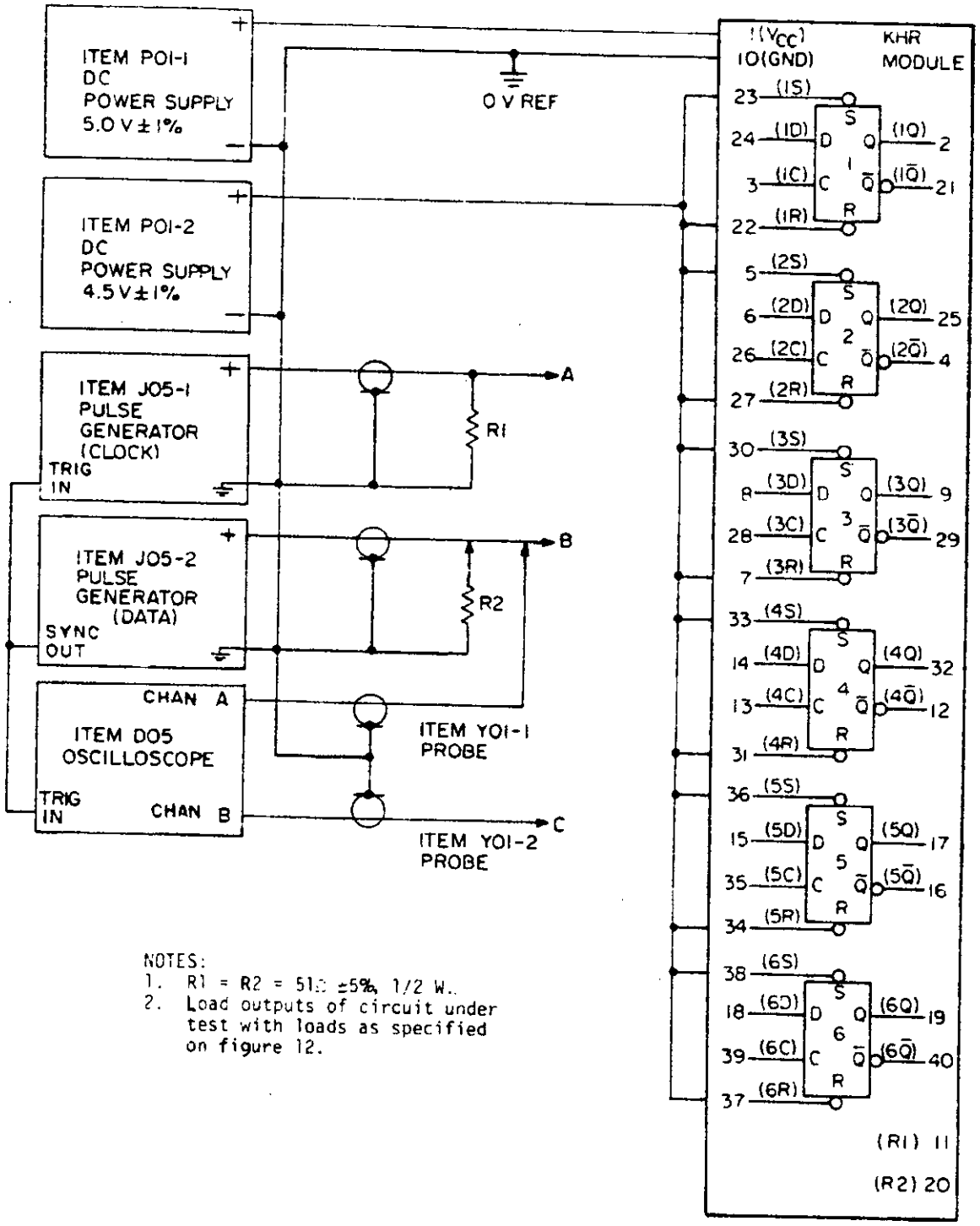
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* TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1Q Propagation delay time (t_{PHL}), clock to output, C_L = 50 pF	3	24	2	2.0	11.5	ns	16,17
Q2	1$\overline{Q}$	3	24	21				
Q3	2Q	26	6	25				
Q4	2$\overline{Q}$	26	6	4				
Q5	3Q	28	8	9				
Q6	3$\overline{Q}$	28	8	29				
Q7	4$\overline{Q}$	13	14	32				
Q8	4$\overline{Q}$	13	14	12				
Q9	5Q	35	15	17				
Q10	5Q	35	15	16				
Q11	6Q	39	18	19				
Q12	6$\overline{Q}$	39	18	40				
Q13	1Q Propagation delay time (t_{PHL}), clock to output, C_L = 150 pF	3	24	2	3	18		
Q14	1$\overline{Q}$	3	24	21				
Q15	2Q	26	6	25				
Q16	2$\overline{Q}$	26	6	4				
Q17	3Q	28	8	9				
Q18	3$\overline{Q}$	28	8	29				
Q19	4Q	13	14	32				
Q20	4$\overline{Q}$	13	14	12				
Q21	5Q	35	15	17				
Q22	5$\overline{Q}$	35	15	16				
Q23	6Q	39	18	19				
Q24	6$\overline{Q}$	39	18	40				

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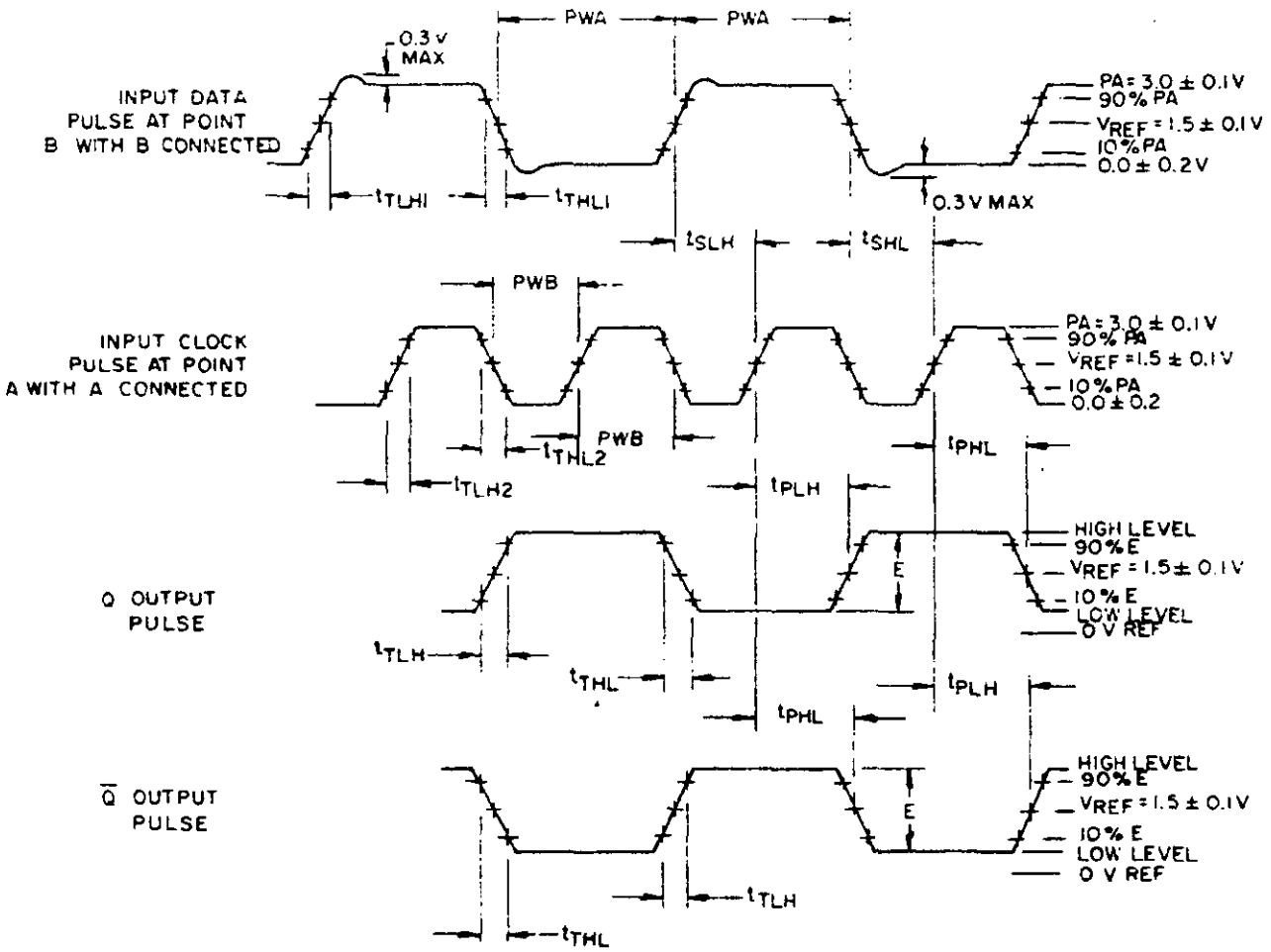
MIL-M-38510/694(EC)



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load outputs of circuit under test with loads as specified on figure 12.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:

   PWA = 500 ±50 ns.
   PWB = 250 ±25 ns.
   $t_{SLH} = t_{SHL}$ = 125 ±10 ns.
   $t_{TLH1} = t_{TLH2} = t_{THL1} = t_{THL2}$ = 4 ±1 ns.

2. Before making the measurements specified in table XIII, verify that the output pulse high lev is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin; shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIII.

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* TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q	Propagation delay time (t_{PLH}), set and reset to output, C_L = 50 pF	23,C	22	A	2	2	8	ns	18,19
R2	1$\bar{Q}$		23	22,C	B	21				
R3	2Q		5,C	27	A	25				
R4	2$\bar{Q}$		5	27,C	B	4				
R5	3Q		30,C	7	A	9				
R6	3$\bar{Q}$		30	7,C	B	29				
R7	4Q		33,C	31	A	32				
R8	4$\bar{Q}$		33	31,C	B	12				
R9	5Q		36,C	34	A	17				
R10	5$\bar{Q}$		36	34,C	B	16				
R11	6Q		38,C	37	A	19				
R12	6$\bar{Q}$		38	37,C	B	40				
R13	1Q	Propagation delay time (t_{PLH}), set and reset to output, C_L = 150 pF	23,C	22	A	2	2.0	17.5		
R14	1$\bar{Q}$		23	22,C	B	21				
R15	2Q		5,C	27	A	25				
R16	2$\bar{Q}$		5	27,C	B	4				
R17	3Q		30,C	7	A	9				
R18	3$\bar{Q}$		30	7,C	B	29				
R19	4Q		33,C	31	A	32				
R20	4$\bar{Q}$		33	31,C	B	12				
R21	5Q		36,C	34	A	17				
R22	5$\bar{Q}$		36	34,C	B	16				
R23	6Q		38,C	37	A	19				
R24	6$\bar{Q}$		38	37,C	B	40				

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* TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Q Propagation delay time (t_{PHL}), set and reset to output, clock low, C_L = 50 pF	23	22,C	B	2	2.0	11.5	ns	18,19
S2	1$\overline{Q}$	23,C	22	A	21				
S3	2Q	5	27,C	B	25				
S4	2$\overline{Q}$	5,C	27	A	4				
S5	3Q	30	7,C	B	9				
S6	3$\overline{Q}$	30,C	7	A	29				
S7	4Q	33	31,C	B	32				
S8	4$\overline{Q}$	33,C	31	A	12				
S9	5Q	36	34,C	B	17				
S10	5$\overline{Q}$	36,C	34	A	16				
S11	6Q	38	37,C	B	19				
S12	6$\overline{Q}$	38,C	37	A	40				
S13	1Q Propagation delay time (t_{PHL}), set and reset to output, clock low, C_L = 150 pF	23	22,C	B	2	2	20		
S14	1$\overline{Q}$	23,C	22	A	21				
S15	2Q	5	27,C	B	25				
S16	2$\overline{Q}$	5,C	27	A	4				
S17	3Q	30	7,C	B	9				
S18	3$\overline{Q}$	30,C	7	A	29				
S19	4Q	33	31,C	B	32				
S20	4$\overline{Q}$	33,C	31	A	12				
S21	5Q	36	34,C	B	17				
S22	5$\overline{Q}$	36,C	34	A	16				
S23	6Q	38	37,C	B	19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:51Z
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* TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

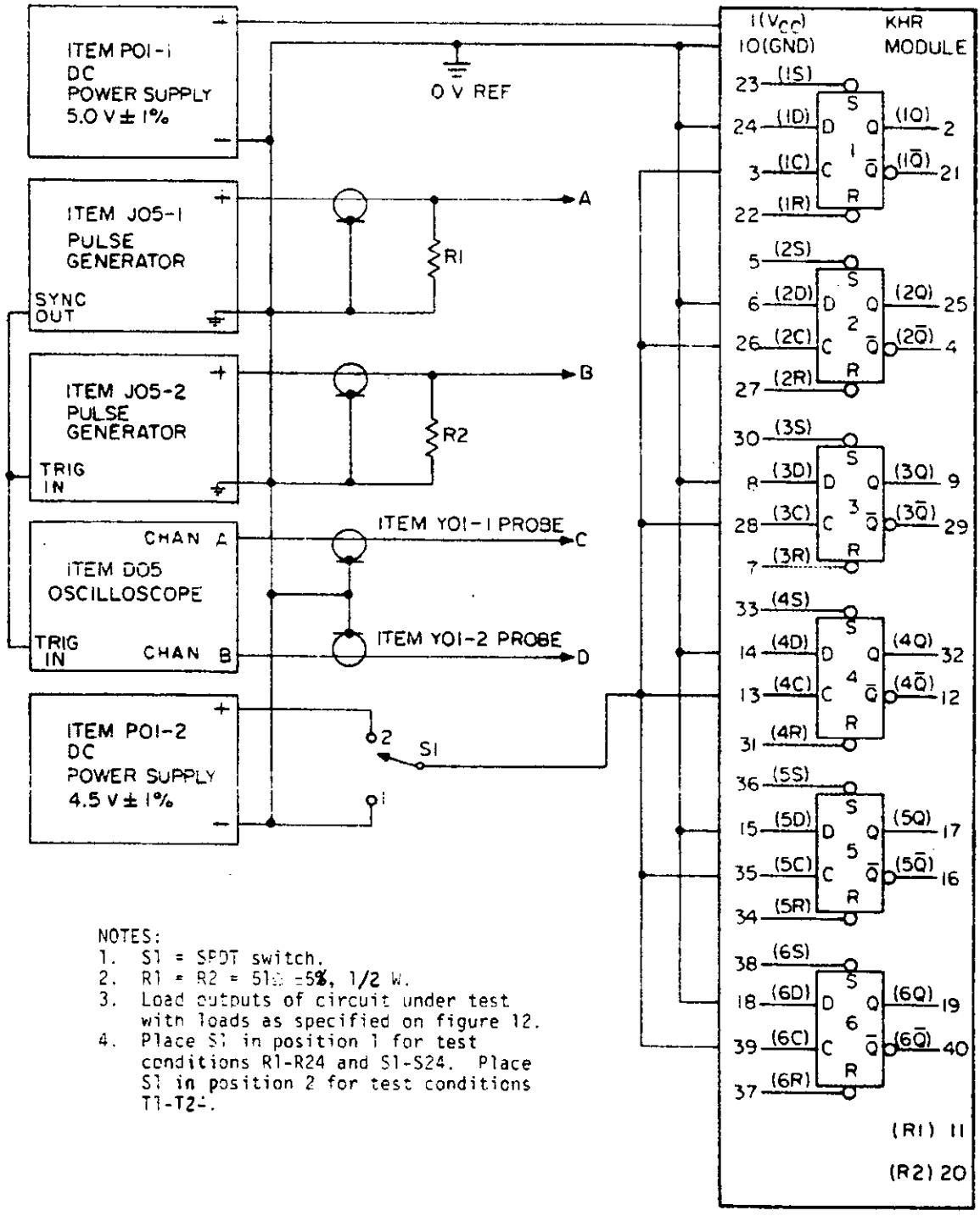
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1Q Propagation delay time (t_{PHL}), set and reset to output, clock high, C_L = 50 pF	23	22,C	B	2	2.0	14.5	ns	18,19
T2	1$\overline{Q}$	23,C	22	A	21				
T3	2Q	5	27,C	B	25				
T4	2$\overline{Q}$	5,C	27	A	4				
T5	3Q	30	7,C	B	9				
T6	3$\overline{Q}$	30,C	7	A	29				
T7	4Q	33	31,C	B	32				
T8	4$\overline{Q}$	33,C	31	A	12				
T9	5Q	36	34,C	B	17				
T10	5$\overline{Q}$	36,C	34	A	16				
T11	6Q	38	37,C	B	19				
T12	6$\overline{Q}$	38,C	37	A	40				
T13	1Q Propagation delay time (t_{PHL}), set and reset to output, clock high, C_L = 150 pF	23	22,C	B	2	2	20		
T14	1$\overline{Q}$	23,C	22	A	21				
T15	2Q	5	27,C	B	25				
T16	2$\overline{Q}$	5,C	27	A	4				
T17	3Q	30	7,C	B	9				
T18	3$\overline{Q}$	30,C	7	A	29				
T19	4Q	33	31,C	B	32				
T20	4$\overline{Q}$	33,C	31	A	12				

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* TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T21	5Q Propagation delay time (t_{PHL}), set and reset to output, clock high, C_L = 150 pF	36	34,C	B	17	2	20	ns	18,19
T22	5Q	36,C	34	A	16				
T23	6Q	38	37,C	B	19				
T24	6$\overline{Q}$	38,C	37	A	40				

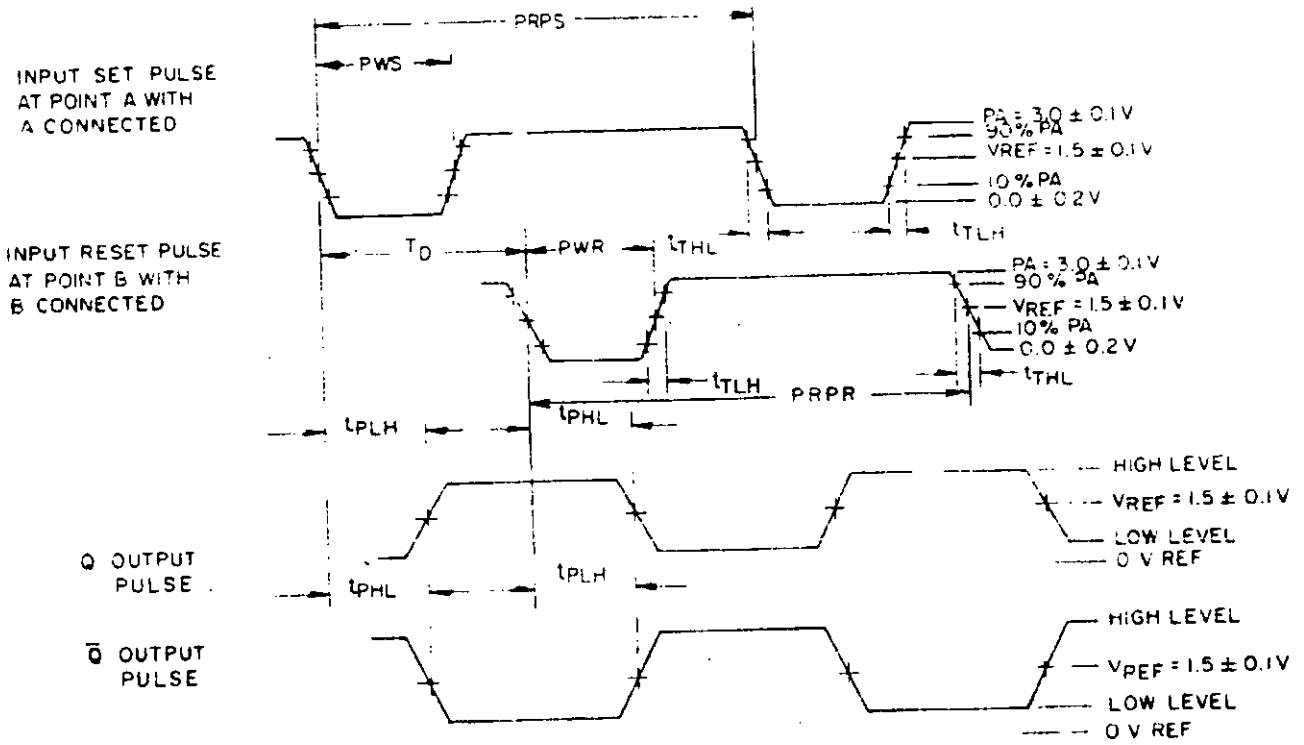
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NOTES:
1. S1 = SPDT switch.
2. R1 = R2 = 51Ω ±5%, 1/2 W.
3. Load outputs of circuit under test with loads as specified on figure 12.
4. Place S1 in position 1 for test conditions R1-R24 and S1-S24. Place S1 in position 2 for test conditions T1-T24.

* FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:51Z
Check the source to verify that this is the current version before use.



NOTES:
1. Adjust items J05-1 and J05-2 to provide input pulse parameters as follows:

   PWS = PWR = 13 ±1 ns.
   PRPS = PRPR = 200 ±20 ns.
   T_D = 100 ±5 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.

2. The lead length of electrical conductors between resistors R1 and R2 and input pins; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin; shall be less than 1 inch (25.4 mm).
3. All voltages are referenced to ground.

★ FIGURE 19. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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* TABLE XV. Setup (t_{SLH} and t_{SHL}) and hold time (t_{HLH} and t_{HHL}) test.

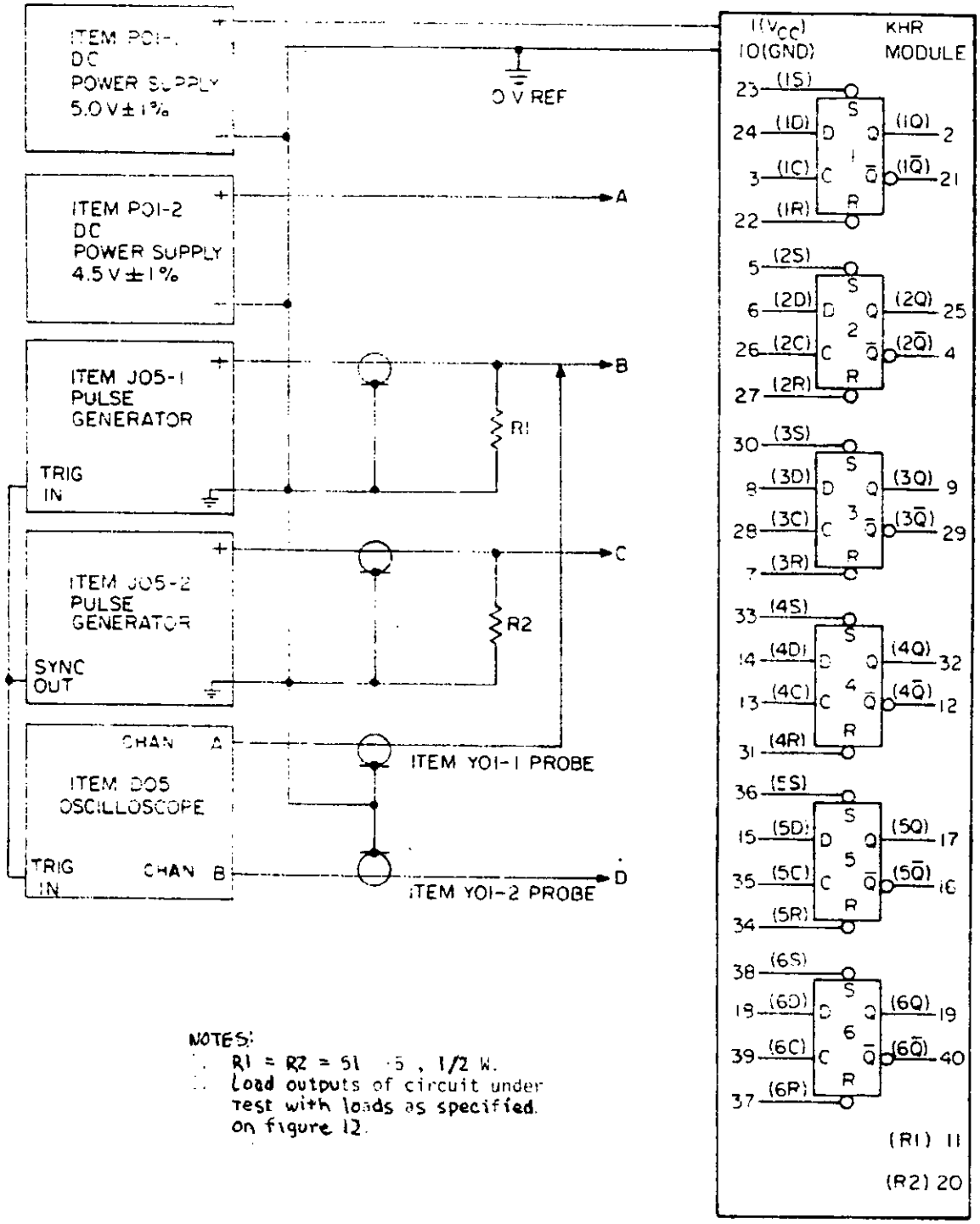
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1D Setup time (t_{SLH}, t_{SHL})	23,22	3	24	2	450	550	ns	20,21
U2	2D	5,27	26	6	25				
U3	3D	7,30	28	8	9				
U4	4D	31,33	13	14	32				
U5	5D	34,36	35	15	17				
U6	6D	37,38	39	18	19				
V1	1D Hold time (t_{HLH}, t_{HHL})	23,22	3	24	2				
V2	2D	5,27	26	6	25				
V3	3D	7,30	28	8	9				
V4	4D	31,33	13	14	32				
V5	5D	34,36	35	15	17				
V6	6D	37,38	39	18	19				

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TABLE XVI. Maximum clock frequency and minimum clock pulse width check test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1C Maximum clock frequency and minimum clock pulse width	23,22	3	24	2	40	48	ns	20,21
W2	2C	5,27	26	6	25	↓	↓	↓	↓
W3	3C	7,30	28	8	9				
W4	4C	31,33	13	14	32				
W5	5C	34,36	35	15	17				
W6	6C	37,38	39	18	19				

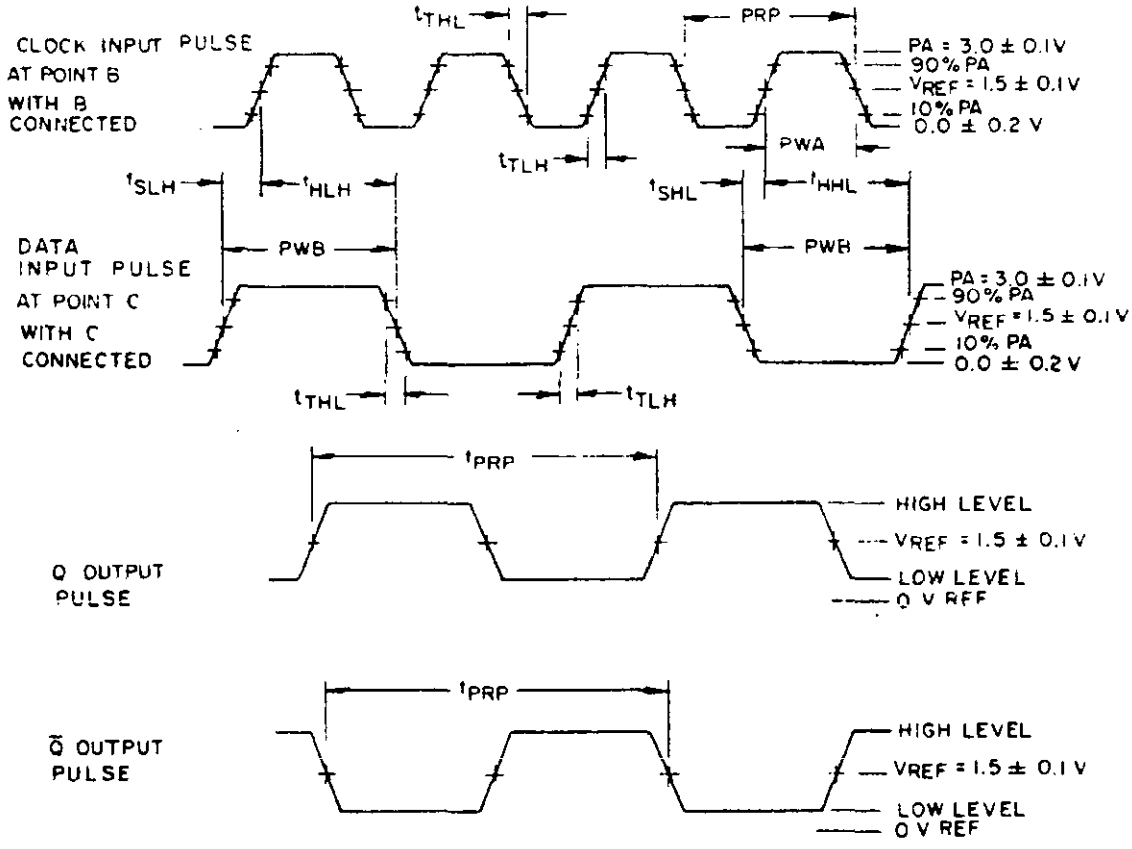
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NOTES:
1. R1 = R2 = 51 ±5 , 1/2 W.
2. Load outputs of circuit under test with loads as specified on figure 12.

* FIGURE 3. Setup (t_{SLH} and t_{SHL}) and hold time (t_{HLH} and t_{HHL}) and maximum clock frequency and minimum clock pulse width check test circuit for tables XV and XVI.

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NOTES:
1. *Input pulse parameters:*

   Clock PRP = PWB = 250 ±25 ns for test conditions U and V.
   Clock PRP = PWB = 22 ±2 ns for test condition W.
   Clock PWA = 125 ±12 ns for test conditions U and V.
   Clock PWA = 13 ±1 ns for test condition W.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.

2. *Adjust data pulse for setup time (t_{SLH} and t_{SHL}) to 3 +2 -0 ns for test condition U.*
3. Adjust data pulse for hold time (t_{HLH} and t_{HHL}) to 2 +2 -0 ns for test condition V.
4. Adjust data pulse for setup time (t_{SLH} and t_{SHL}) to 10 ±1 ns for test condition W.

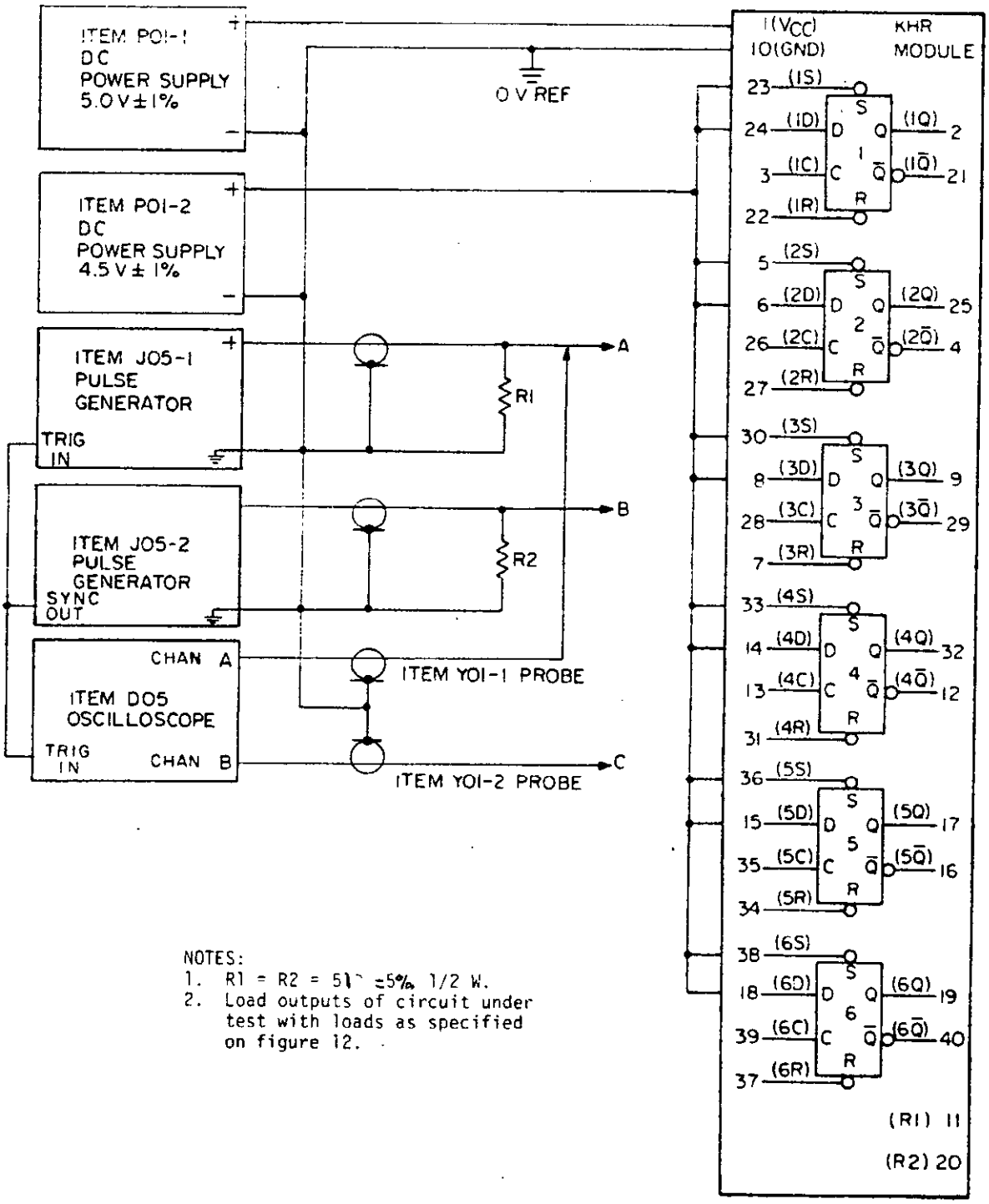
* FIGURE 21. Setup (t_{SLH} and t_{SHL}) and hold time (t_{HLH} and t_{HHL}) and maximum clock frequency and minimum clock pulse width check test circuit waveforms for tables XV and XVI.

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TABLE XVII. Reset recover time test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1R Reset recovery time	3	22	2	450	550	ns	22,23
X2	2R	26	27	25				
X3	3R	28	7	9				
X4	4R	13	31	32				
X5	5R	35	34	17				
X6	6R	39	37	19				

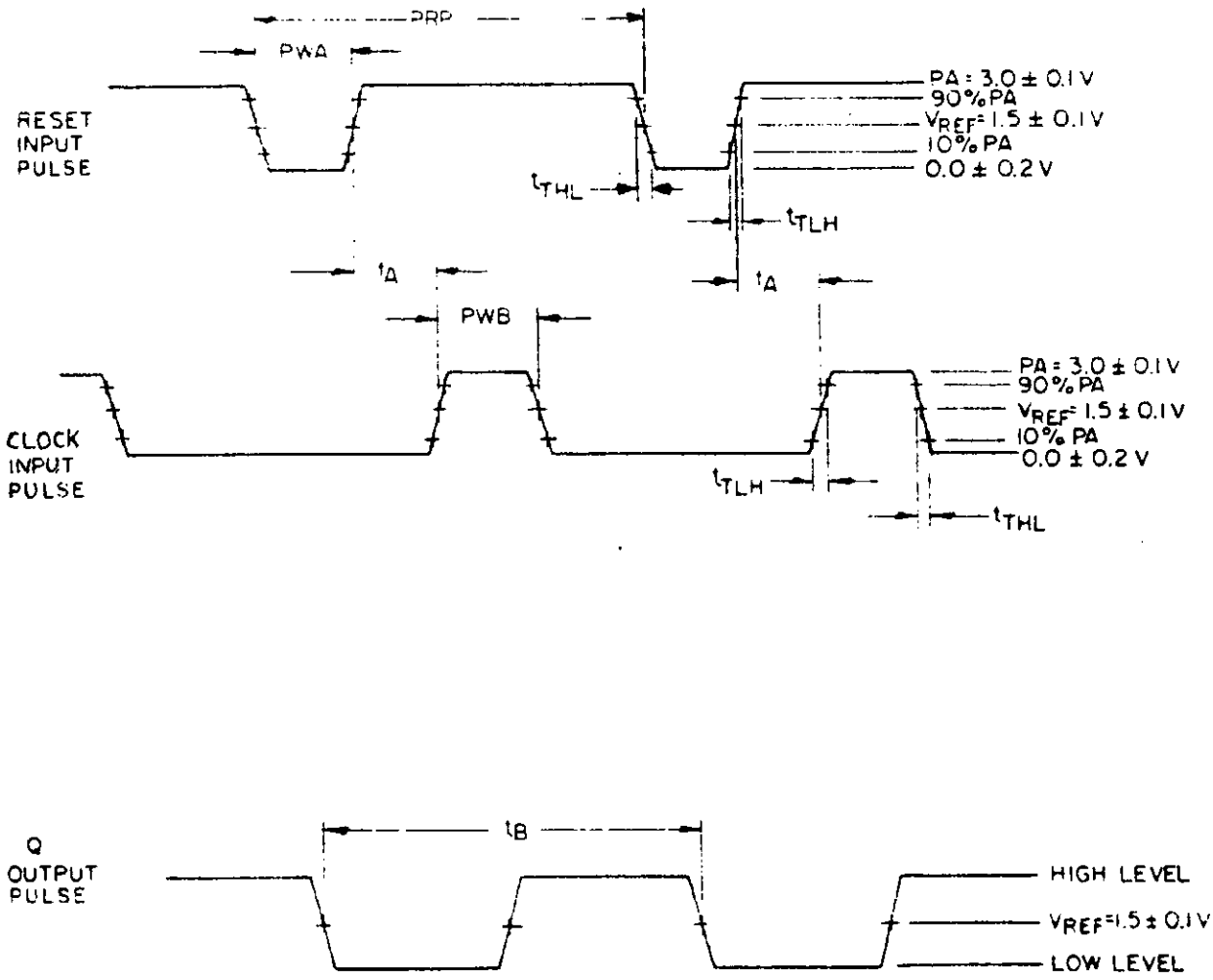
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load outputs of circuit under test with loads as specified on figure 12.

* FIGURE 22. Reset recovery time test circuit for table XVII.

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NOTES:
1. Input pulse parameters:

   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = PWB = 100 ±10 ns.
   PRP = 500 ±60 ns.

2. Set reset recovery time t_A to 13 ±1 ns.

* FIGURE 23. Reset recovery time test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level and high-level) test.

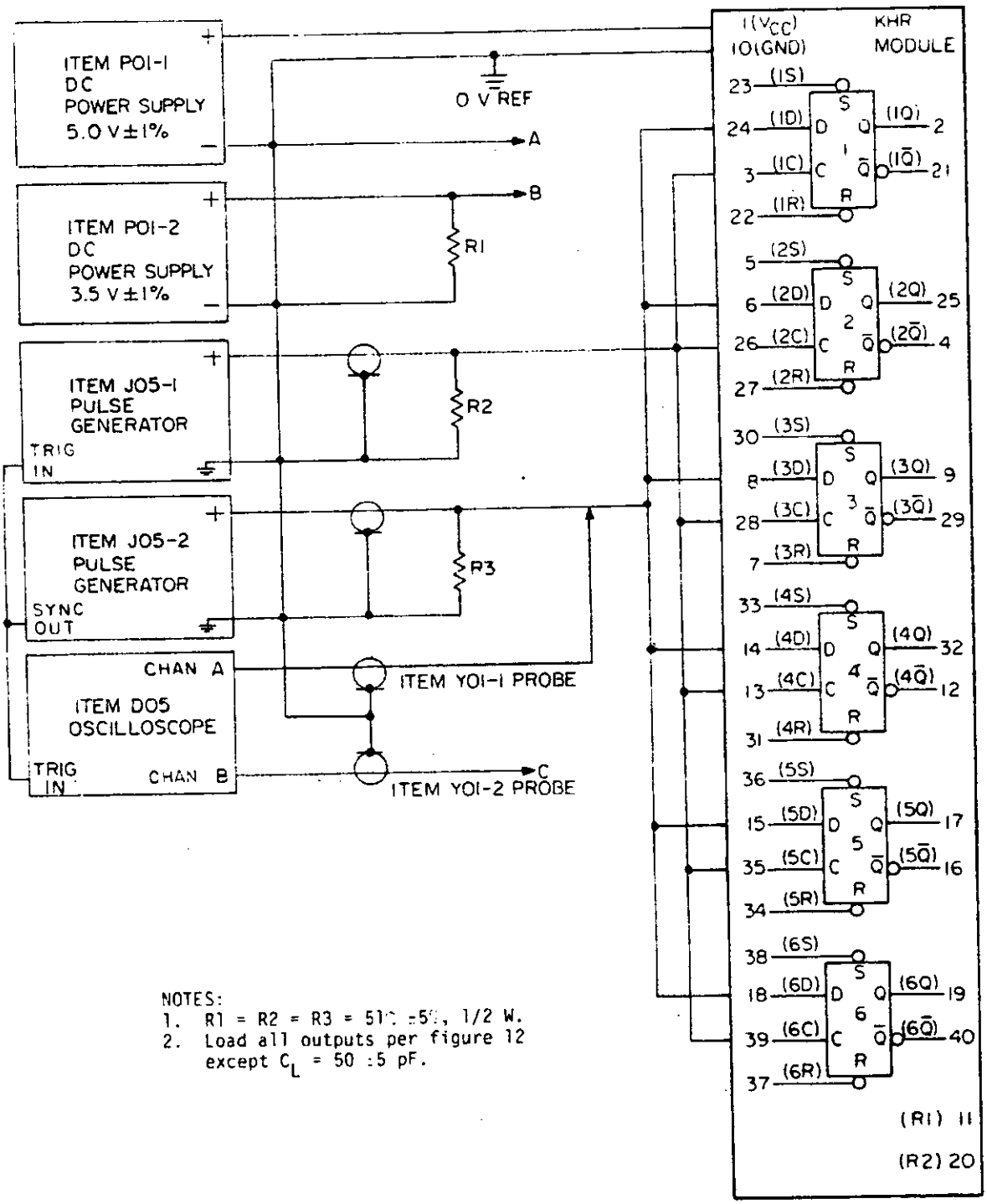
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Y1	1Q Output crosstalk, low level	22	23,27,5,7,30,31,33,34,36,37, 38	2	0	500	mV	24,25
Y2	1$\overline{Q}$	23	22,27,5,7,30,31,33,34,36,37, 38	21				
Y3	2Q	27	22,23,5,7,30,31,33,34,36,37, 38	25				
Y4	2$\overline{Q}$	5	22,23,27,7,30,31,33,34,36, 37,38	4				
Y5	3Q	7	22,23,27,5,30,31,33,34,36, 37,38	9				
Y6	3$\overline{Q}$	30	22,23,27,5,7,31,33,34,36,37, 38	29				
Y7	4Q	31	22,23,27,5,7,30,33,34,36,37, 38	32				
Y8	4$\overline{Q}$	33	22,23,27,5,7,30,31,34,36,37, 38	12				
Y9	5Q	34	22,23,27,5,7,30,31,33,36,37, 38	17				
Y10	5$\overline{Q}$	36	22,23,27,5,7,30,31,33,34,37, 38	16				
Y11	6Q	37	22,23,27,5,7,30,31,33,34,36, 37	19				
Y12	6$\overline{Q}$	38	22,23,27,5,7,30,31,33,34,36, 37	40				

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TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	1Q Output crosstalk, high level	23	22,27,5,7,30,31,33,34,36,37, 38	2	0	400	mV	24,25
Z2	1$\bar{Q}$	22	23,27,5,7,30,31,33,34,36,37, 38	21				
Z3	2Q	5	22,23,27,7,30,31,33,34,36, 37,38	25				
Z4	2$\bar{Q}$	27	22,23,5,7,30,31,33,34,36,37, 38	4				
Z5	3Q	30	22,23,27,5,7,31,33,34,36,37, 38	9				
Z6	3$\bar{Q}$	7	22,23,27,5,30,31,33,34,36, 37,38	29				
Z7	4Q	33	22,23,27,5,7,30,31,34,36,37, 38	32				
Z8	4$\bar{Q}$	31	22,23,27,5,7,30,33,34,36,37, 38	12				
Z9	5Q	36	22,23,27,5,7,30,31,33,34,37, 38	17				
Z10	5$\bar{Q}$	34	22,23,27,5,7,30,31,33,36,37, 38	16				
Z11	6Q	38	22,23,27,5,7,30,31,33,34,36, 37	19				
Z12	6$\bar{Q}$	37	22,23,27,5,7,30,31,33,34,36, 38	40				

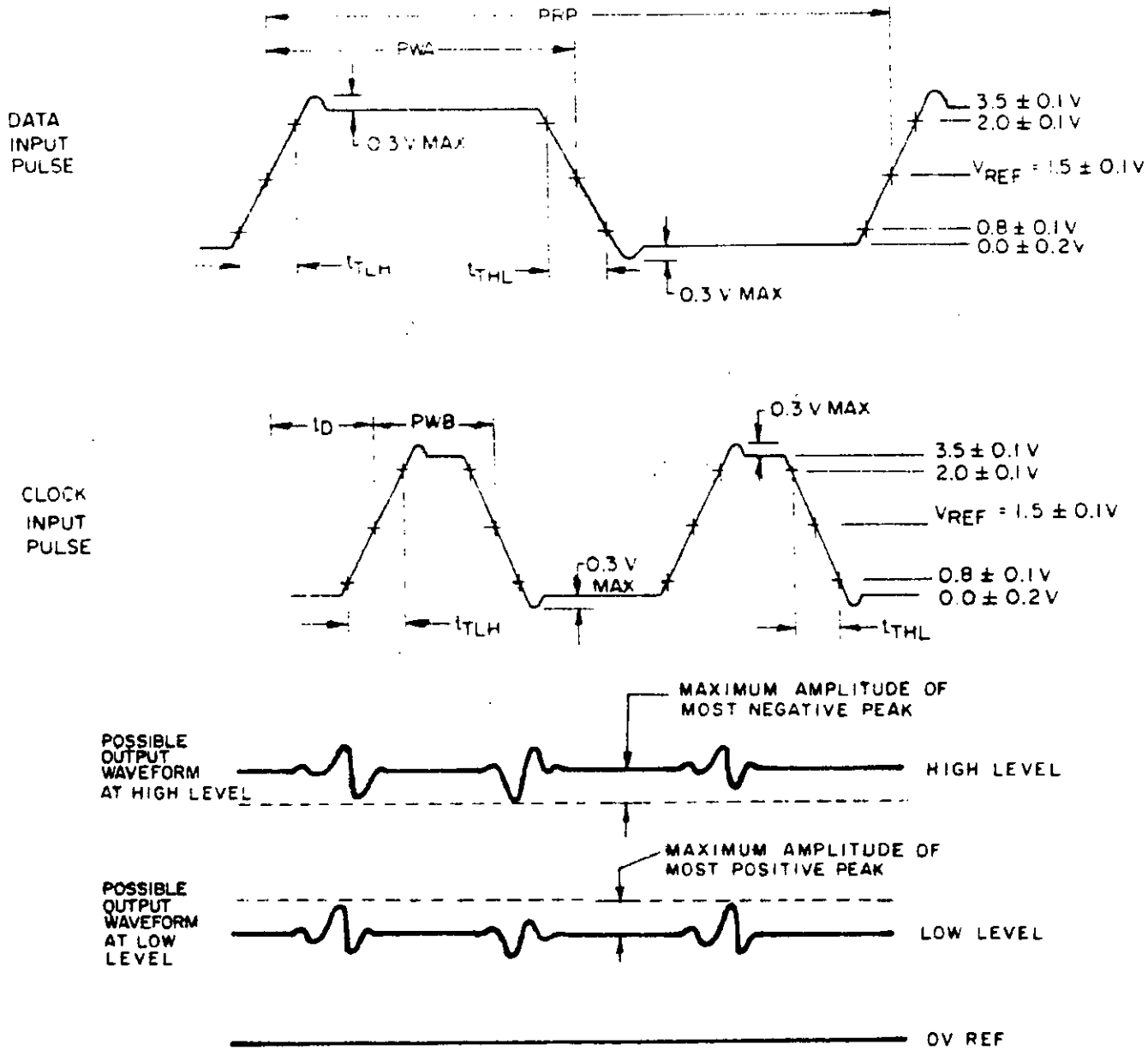
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NOTES:
1. R1 = R2 = R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 12 except C_L = 50 ±5 pF.

* FIGURE 24. Output crosstalk (low-level and high-level) test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:

   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   PWB = 100 ±10 ns.
   $t_D = 200 \pm 20$ ns.

2. The lead length of electrical conductors between resistors R2 and R3 and input pins; between probes YO1-1 and YO1-2 and input and output pins; between load circuits and output pins; and between ground leads of probes YO1-1 and YO1-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

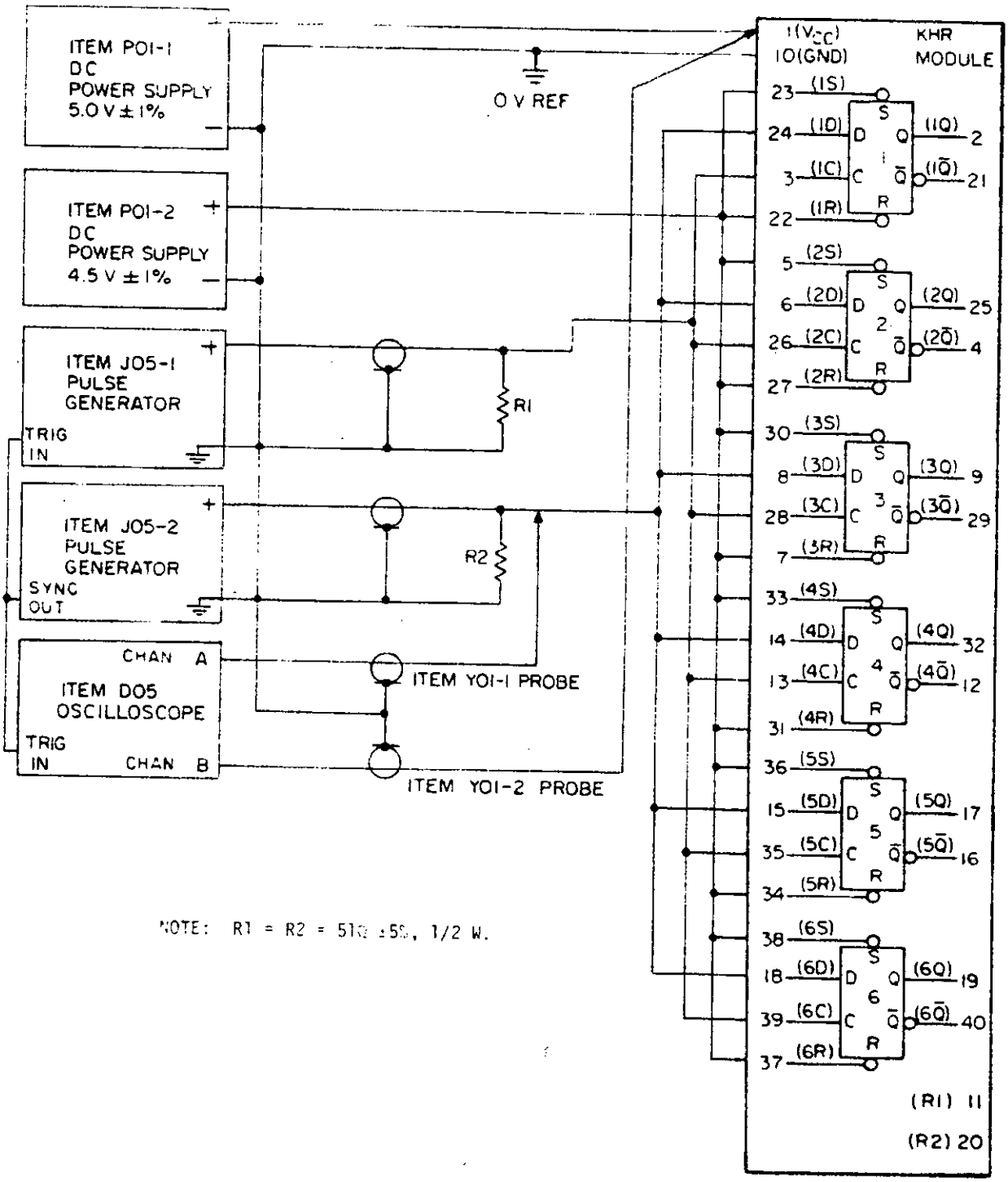
FIGURE 25. Output crosstalk (low-level and high-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
AA	Power supply transient amplitude	0	400	mV	26,27

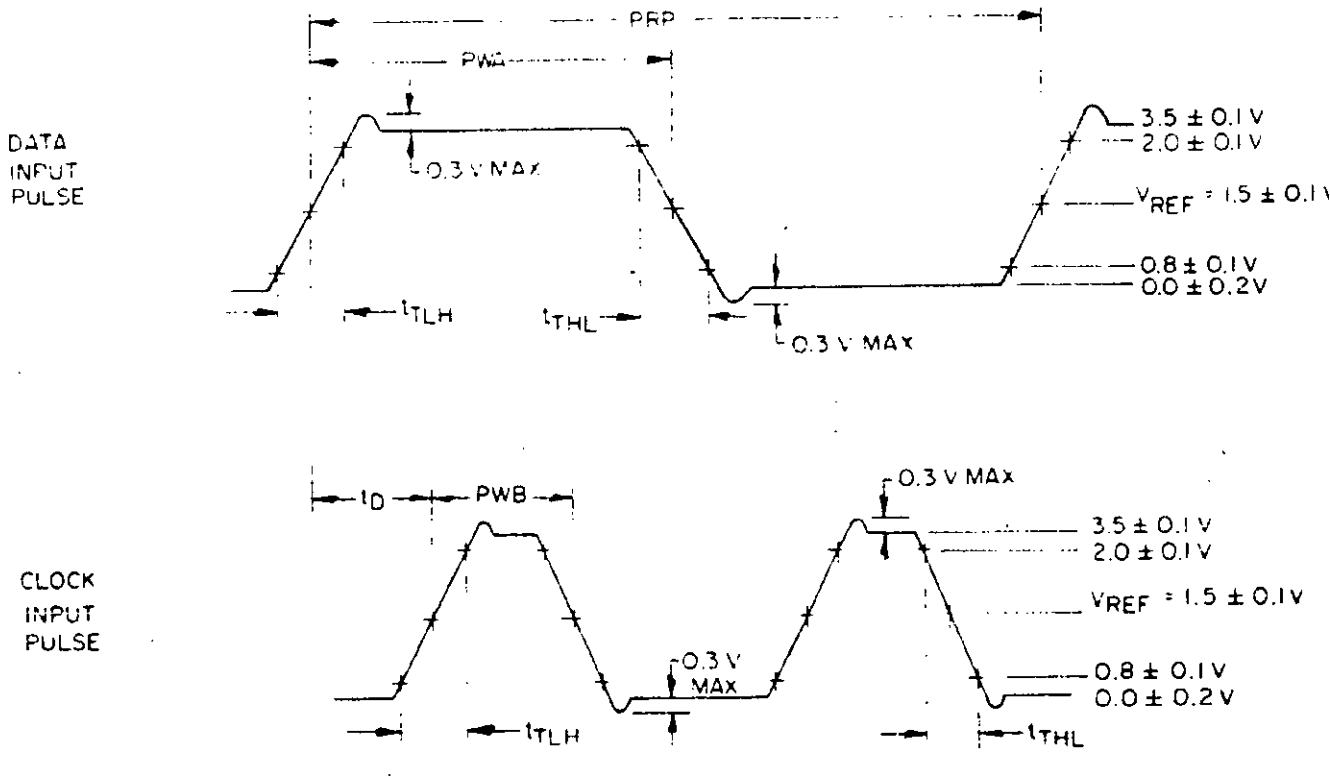
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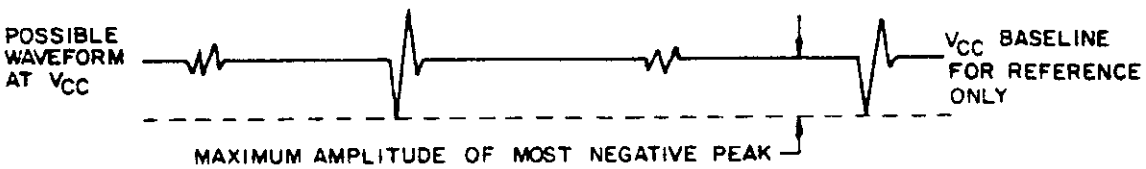


NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 26. Power supply transient amplitude test circuit for table XIX.

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NOTE: Input pulse parameters:

$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PWA = 500 ±50 ns.
PWB = 100 ±10 ns.
PRP = 1,000 ±100 ns.
t_D = 200 ±20 ns.

FIGURE 27. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB	Power supply filter capacitance	Open	1	1.6	3.0	µF	28

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TABLE XXI. Input capacitance test.

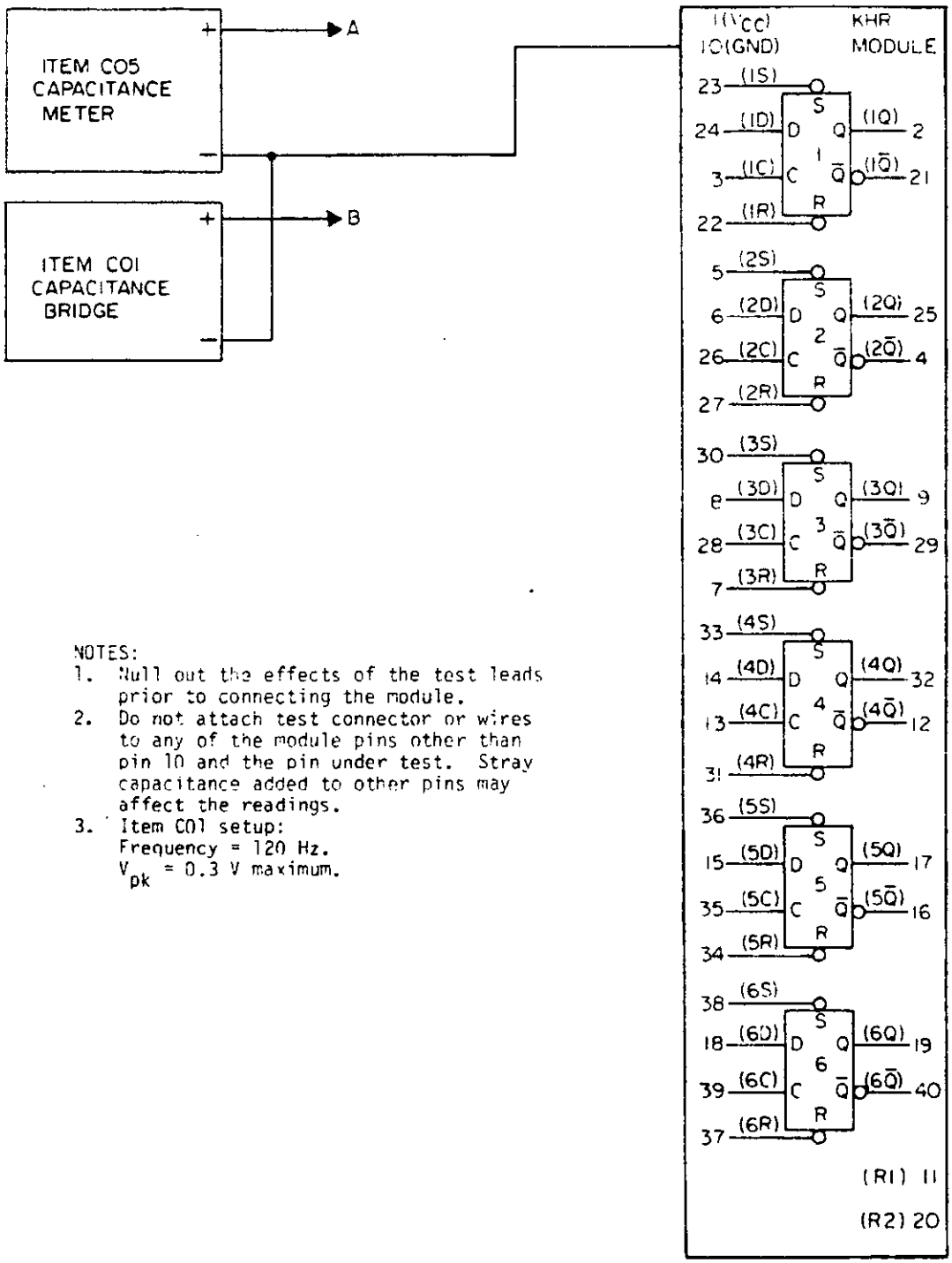
Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
CC1	1D	Input capacitance	24	Open	2	15	pF	28
CC2	2D		6					
CC3	3D		8					
CC4	4D		14					
CC5	5D		15					
CC6	6D		18					
CC7	1C		3					
CC8	2C		26					
CC9	3C		28					
CC10	4C		13					
CC11	5C		35					
CC12	6C		39					
CC13	1S		23					
CC14	2S		5					
CC15	3S		30					
CC16	4S		33					
CC17	5S		36					
CC18	6S		38					

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TABLE XXI. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC19	1R Input capacitance	22	Open	2	15	pF	28
CC20	2R	27					
CC21	3R	7					
CC22	4R	31					
CC23	5R	34					
CC24	6R	37					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item CO1 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 28. Power supply filter capacitance and input capacitance test circuit for tables XX and XXI.

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